2025 Graduate Thesis

Deep-Q-Network-Based Optimization Framework for Diversion Using Real Flight Data in Airport Closure

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Chapter 1
Introduction

1. Introduction

1.1 Background

In recent years, the aviation industry has witnessed a growing number of airport disruptions caused by pilot errors, mechanical failures, severe weather, and runway incursions. For instance, on January 2nd, 2024, a ground collision at Tokyo Haneda Airport between Japan Airlines Flight 516 and a Japan Coast Guard aircraft led to a complete runway closure for over 3 hours. Dozens of incoming aircraft were forced to divert, causing cascading operational impacts. Similar diversion-inducing events include Jeju Air Flight 2216's overrun at Muan Airport (2024) and Air India Flight 171's accident at Sardar Vallabhbhai Patel International (Ahmedabad) Airport (2025).

Such accidents highlight the increasing complexity of aircraft diversion decisions, which must account for factors such as remaining fuel, aircraft position, weather conditions, and destination capacity. However, these decisions remain heavily reliant on the judgment of pilots and air traffic controllers, often based on static or delayed information, leading to suboptimal choices and operational inefficiencies.

These cases illustrate how diversion decisions must be made under intense time pressure, with incomplete situational awareness and multiple competing factors—remaining fuel, aircraft location, en-route weather, available runway length, and airport capacity. In many cases, pilots and air traffic controllers rely on pre-established diversion lists and voice communication, which may not reflect the latest operational constraints or weather changes. This reactive and manual decision-making process can result in longer diversion distances, unnecessary holding patterns, or congestion at certain alternate airports, ultimately reducing the overall resilience and efficiency of the air traffic system.

1.2 Previous work

Research on airline disruption management provides comprehensive models for aircraft, crew, and passenger recovery, but most studies focus on network/schedule recovery after a diversion decision has already been made rather than on the real-time selection of the diversion airport itself [1][2]. Recent reviews also highlight practical limitations—information latency, fragmented situational awareness across ATC/airline operations, and decision-time pressure—and point to gaps in jointly optimizing operational efficiency and passenger impact at the moment of choice [1][2].

In parallel, the aircraft trajectory optimization literature typically minimizes fuel (or fuel–time composites) under performance and procedural constraints. Physics-based methods rely on EUROCONTROL BADA to keep candidate trajectories within feasible thrust/drag/SFC envelopes

with consistent mass updates [3]. For realism, high-fidelity reanalysis meteorology such as ERA5 is interpolated along discretized paths to estimate winds, temperature, and density that drive fuel flow and timing [4]. Recent studies further explore machine-learning-assisted trajectory optimization under realistic constraints, reinforcing that trajectory-level fuel/time trade-offs are central for comparative evaluation [5].

More recently, reinforcement learning (RL) has been applied to air-traffic/aviation tasks—conflict resolution, demand–capacity balancing, and real-time routing—showing that deep RL (including DQN variants) can learn policies under complex, high-dimensional dynamics, often in multi-agent settings. Surveys and applications report promising performance while also noting generalization and certification challenges in safety-critical contexts [6][7][8]. Notably, while DRL-based routing/contingency studies demonstrate multi-objective trade-offs, most do not couple policy selection with a physics-based trajectory optimizer that quantifies fuel/time feasibility under aircraft-performance and meteorological constraints [6][7][8].

Scope relative to prior work. A gap therefore remains between the selection of a diversion airport and the trajectory-level feasibility of that choice. This thesis addresses the gap by linking a policy selector (DQN) with a direct, physics-based trajectory optimizer so that diversion policies can be evaluated on time and fuel at the trajectory level.

Chapter 2
Problem statement

2. Problem statement

2.1 Current process of decision making for division

In the event of a runway closure caused by an accident at an airport, the diversion decision-making process follows a multi-step operational protocol. [9] For instance, when a closure occurs at Haneda Airport, the actual procedure unfolds as follows:

1. The control tower at Haneda promptly notifies the responsible Area Control Center (ACC) and the Air Traffic Management Center (ATMC) of the runway closure.
2. The ACC then informs en-route aircraft at the time of the accident, while the ATMC compiles a list of available landing slots at alternative airports and provides this information to each airline.
3. Based on the provided data, airlines analyze feasible diversion airports, considering factors such as slot availability, weather, and proximity, and provide this information to the flight crew.
4. Pilots evaluate candidate airports by considering fuel constraints, weather, distance to destination, and passenger convenience, and select a diversion airport, reporting their decision back to the ACC. [10]
5. The ACC sets up a new route to the selected diversion airport and guides the aircraft safely through the revised flight path.

This process is illustrated in figure 2-1 below, which depicts the current diversion decision-making flow used in Japanese airspace.

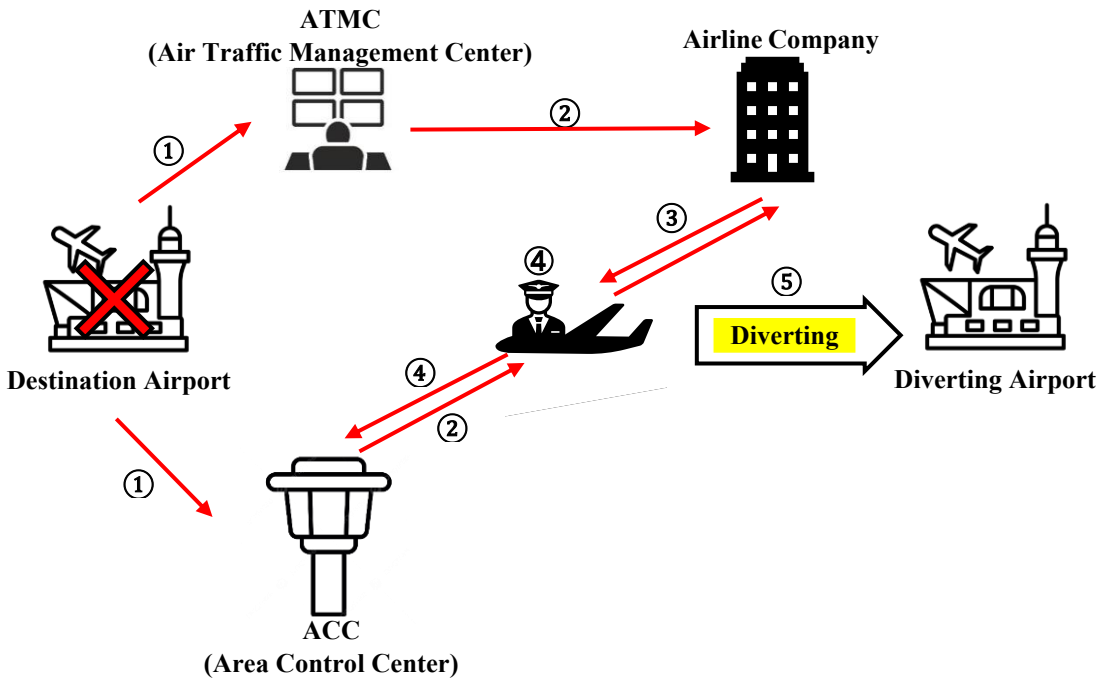


Figure 2-1 Actual Diversion Decision Process in Practice

However, in actual operations, a critical challenge arises: multiple aircraft often select the same diversion airport simultaneously based on identical information, leading to sudden congestion. As a result, landing slots become insufficient, holding patterns lengthen, and fuel exhaustion risks increase. These disruptions further cause inconvenience for passengers, arrival delays, and cascading effects on subsequent flights.

This analysis is based on operational insights gathered from interviews with Dr. Yasutaka Kawamoto (a pilot at Japan Airlines) and Mr. Takumi Ishida (a controller at Tokyo Approach Center).

2.2 Research objective

The current aircraft diversion decision-making process faces several limitations, including vulnerability to congestion, a lack of predictive capability, and the absence of real-time optimization. This research aims to address these issues by proposing an intelligent diversion decision-making model that enhances the resilience of the air traffic system in the event of airport closures or other operational disruptions.[11]

Unlike the traditional approach, which relies on static and manual information exchanges between stakeholders, this study employs reinforcement learning based on a Deep Q-Network (DQN) to optimize aircraft diversion decisions.[12] A learning environment is constructed by integrating actual

flight operational data, airport slot availability, weather conditions, and aircraft performance models, so that the agent can select the optimal alternate airport under various operational scenarios.

Rather than merely mimicking human decisions, the proposed model is designed to make timely and effective decisions in complex, time-constrained situations where human responses may be delayed due to procedural bottlenecks. By incorporating fuel efficiency and passenger convenience into its optimization scheme, the model seeks not only to match but also to enhance decision quality under pressure. Its effectiveness will be evaluated by comparing DQN-selected outcomes with actual diversion data during Haneda airport closure on January 2nd 2024. This research ultimately contributes to the development of a data-driven, real-time decision-making framework that enhances the resilience and agility of air traffic operations in response to unforeseen disruptions. [13]

Chapter 3
Method

3. Method

3.1 Research framework

This study proposes a comprehensive framework for optimizing aircraft diversion decisions under airport closure scenarios. The overall process is designed to integrate actual flight data, aircraft performance models, fuel estimation, reinforcement learning, and comparative evaluation with actual operations.

The process begins with data collection from Flightradar24 [14], targeting all inbound flights to Tokyo International Airport (Haneda Airport) on January 2nd, 2024, the day of the ground collision accident happened. A filtering step ensures that only complete and valid flight trajectories are retained, resulting in 59 selected flights for analysis (i.e., flights scheduled to arrive between 08:47 and 12:30 UTC, with no significant data gaps).

For each of these flights, a fuel estimation tool—developed using the BADA aircraft performance model [3] and numerical prediction meteorological data [15]—is used to estimate the remaining fuel at the time of the accident. [16] This estimation is critical for determining the feasibility of diversion options.

Subsequently, a flight trajectory optimization tool calculates the fuel consumption required to divert from the aircraft's position at the time of the accident to each candidate airport. These results form the input for the decision-making agent.

Using these estimates, a Deep Q-Network (DQN) is trained to optimize the selection of a diversion airport. The model learns a decision-making policy that maps the aircraft's real-time state (e.g., location, fuel, congestion) to the optimal diversion airport under a reward structure prioritizing fuel efficiency, passenger convenience, and total flight time. The agent explores various scenarios and learns the policy that maximizes the overall reward.

Finally, the trained model is evaluated through simulation, and its decisions are compared to the actual diversion outcomes recorded on the day of the accident. The comparison focuses on key metrics such as fuel efficiency and passengers convenience.

Figure 3-1 illustrates the overall workflow of the proposed method, from data collection through simulation and evaluation.

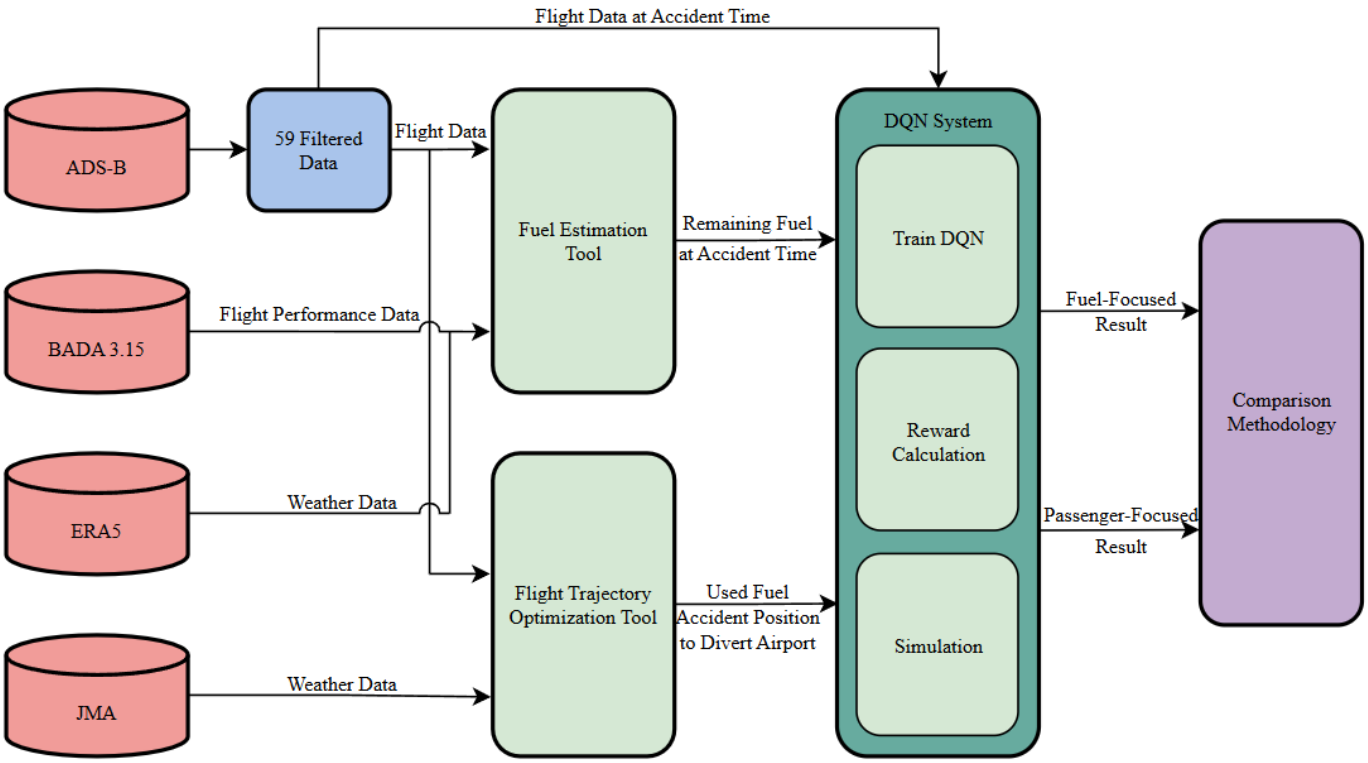


Figure 3-1 Overall Flow of Research

3.2 Deep Q-Network (DQN) based diversion optimization method

3.2-1 Reason for adopting DQN in diversion decision-making

In emergency scenarios such as sudden airport closures, pilots must make fast and accurate decisions about which alternate airport to divert to. Traditionally, this decision depends on pre-selected alternate airports in the flight plan (usually 1 to 2), limited real-time communication with airline operations control, and incomplete information about current aircraft states and airport slot availability.

However, in reality, much larger set of potential diversion combinations exists. In our specific problem setup, considering 59 aircraft with 5 alternate airports, the total number of possible combinations is approximately:

$$5^{59} = 1.73 \cdot 10^{41}$$

This is a combinatorial explosion far beyond human cognitive or procedural capability. Pilots, under time pressure and uncertainty, cannot be expected to evaluate such large decision spaces. Furthermore, factors such as remaining fuel, diversion distance, and arrival slot congestion must be jointly considered — making the problem a multi-objective combinatorial optimization task under constraints.

To efficiently handle this, I employ a Deep Q-Network (DQN), a reinforcement learning method well-suited for decision-making in discrete action spaces. The DQN framework can learn from past flight data to infer optimal diverting actions in real time, while rapidly evaluating trade-offs between competing objectives such as fuel efficiency, arrival feasibility, and distance from the original destination. Unlike traditional methods, DQN enables:

- Scalability to hundreds of simultaneous aircraft
- Real-time decisions based on continuous input state (latitude, longitude, speed, fuel, etc.)
- Learning from data instead of hand-crafted rules
- Adaptation to different objectives, such as airlines' cost-efficiency or passengers' convenience

3.2-2 Structure of the system and operational workflow

The DQN-based diversion decision system consists of modular components operating in a closed-loop reinforcement learning (RL) framework. The system enables real-time and adaptive decision-making for each aircraft by combining state observation, policy inference, simulation, and reward evaluation.

The entire workflow is summarized in Figure 3-2, and the core components are outlined in Table 3-1. The system was implemented in Python 3.10 using PyTorch (DQN agent; adapted from the official PyTorch DQN tutorial), NumPy/SciPy (numerics), pandas (data handling), Matplotlib (plots), xarray/netCDF4 (meteorological data I/O), and pyproj/cartopy/shapely (geodesy & map projection)

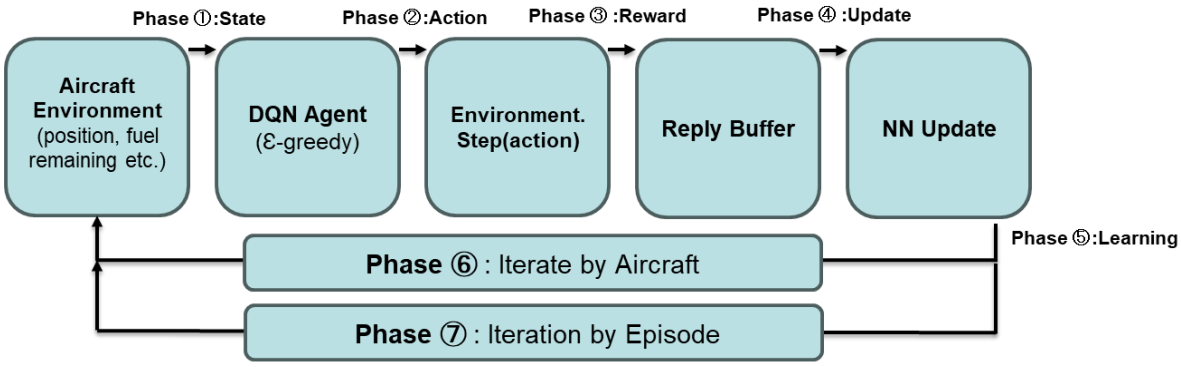


Figure 3-2 Operational flow of DQN-based aircraft diverting system

Table 3-1 Components of DQN-based Diversion System

Component	Description
Aircraft Environment	Extracts the aircraft's current state vector (lat, lon, altitude, fuel, etc.)
DQN Agent	Chooses a diversion airport using an ε-greedy policy based on Q-value estimation
Environment Step	Simulates the action (e.g., fuel feasibility, slot congestion) and calculates multi-factor reward
Replay Buffer	Stores past transitions for off-policy training
NN Update	Updates the Q-network using sampled minibatches and MSE loss
Phase 6: Iteration by Aircraft	For each episode, repeat decision steps across all aircraft
Phase7: Iteration by Episode	Repeat the entire aircraft iteration over multiple episodes (e.g., 2000)

State definition (aircraft environment)

Each aircraft is represented by a six-dimensional state vector derived as shown below from ADS-B and fuel estimation data at the time of the accident (08:47 UTC):

[latitude, longitude, distance to Haneda (km), altitude (m), ground speed (m/s), remaining fuel (kg)]

These states are stored as real_flight_states_with_position_final.csv and are used as input to the DQN agent. Additional data includes flight ID, aircraft type, and origin coordinates.

Action space and environment simulation

In the proposed DQN-based diverting framework, the agent selects one of five candidate diversion airports: Narita (NRT), Nagoya (NGO), Kansai (KIX), Osaka Itami (ITM), or Sendai (SDJ). These airports were selected based on the actual diversion patterns observed on January 2nd, 2024, during the Haneda Airport accident, focusing on large airports and those most frequently used as diversion destinations on that day. Each airport is defined not only by its geographic coordinates but also by specific operational constraints such as slot availability and empirically determined bonus values that differ depending on the evaluation perspective (e.g., passenger convenience or airline efficiency).

Once an action (i.e., a diversion airport) is selected, the environment module, RerouteEnv, simulates the outcome of this decision using a step function. The simulation first verifies feasibility by comparing the fuel required for diverting with the aircraft's remaining fuel. It then checks for available landing slots at the estimated arrival time based on a slot schedule that accounts for dynamic airport usage.

The agent's decision is evaluated through a reward function composed of multiple criteria: proximity to the original destination (Haneda Airport), fuel efficiency of the selected route, penalties for congestion based on slot utilization, and an additional penalty if the aircraft attempts to return to its origin airport. If a chosen action is invalid—due to excessive fuel requirements or lack of available slots—the environment assigns a large negative reward. Slot-based penalties are incrementally scaled according to how saturated each airport becomes over the simulation, encouraging the agent to balance between feasibility and operational efficiency.

Replay buffer and network update

Each interaction between the agent and the environment is stored as a transition tuple:

$$(s, a, r, s')$$

where s is the current state, a is the selected action, r is the received reward, and s' is the resulting next state.

These transitions are stored in a replay buffer with a capacity of 10,000 entries. Once sufficient samples are collected, the system draws random minibatches (size: 32) to update the Q-network. The loss is computed as the Mean Squared Error (MSE) between the predicted Q-values and the target Q-values based on the Bellman equation.

This experience replay mechanism decouples learning from sequential action execution, thereby improving sample efficiency and training stability. The Q-network is optimized using the Adam optimizer with a learning rate of 0.001, and a discount factor γ=0.95. The exploration rate ϵ decays gradually over episodes to ensure a smooth transition from exploration to exploitation.

The hyperparameter settings are detailed in Appendix A.

Training loop: phased execution

In my implementation, the diversion decision-making workflow is divided into sequential phases. From Phases 1 to 5 encompass data preprocessing, environment initialization, and initial model setup, preparing the system for the iterative training process described in the following section.

The workflow is organized into two key iterative loops as shown:

- Phase 6: Iteration by Aircraft
  For each training episode, the system iterates over all aircraft in the dataset. Each aircraft undergoes a reset of its environment, and a single step is executed to collect reward data.
- Phase 7: Iteration by Episode
  The full process is repeated across multiple episodes (typically 2000), with exploration rate ε decaying over time. This iterative approach enables the model to generalize better and refine its diverting policy across various aircraft conditions.

3.2-3 Reward and penalty formulation

To emulate realistic decision-making under emergency diversion scenarios, the reward function in this study was carefully designed to reflect both perspectives of airlines and passengers. The DQN agent learns to choose a diversion airport from multiple candidate airports based on a composite reward function comprising operational feasibility, geographic proximity, passenger inconvenience, and logistical constraints like airport congestion and fuel availability.

This section presents a detailed breakdown of each reward model, the components involved, the logic behind penalties, and how dynamic elements such as return-to-origin were integrated.

Common reward structure

The reward is composed of four core components as shown in Table 3-2, each reward is normalized and scaled to ensure consistent magnitude during training. These components are common to both fuel- and passenger-focused models.

Table 3-2 Reward Structure and Formula

Component	Description	Formula or Method
dist_divert_score	Proximity to the chosen diverting airport	$\exp\left(-d_{dist_{divert}}/250\right)$
dist_HND_score	Proximity to the original destination airport (Haneda Airport)	$\exp\left(-d_{dist_{HND}}/450\right)$
origin_score	Penalty for returning to departure airport (encourages forward progress)	$\frac{1}{1+\exp\left(\left(d_{origin}-500\right)/150\right)}$
fuel_score	Fuel efficiency based on estimated fuel consumption to reach diverting airport	$\exp\left(-\text{fuel}_{used}/4500\right)$

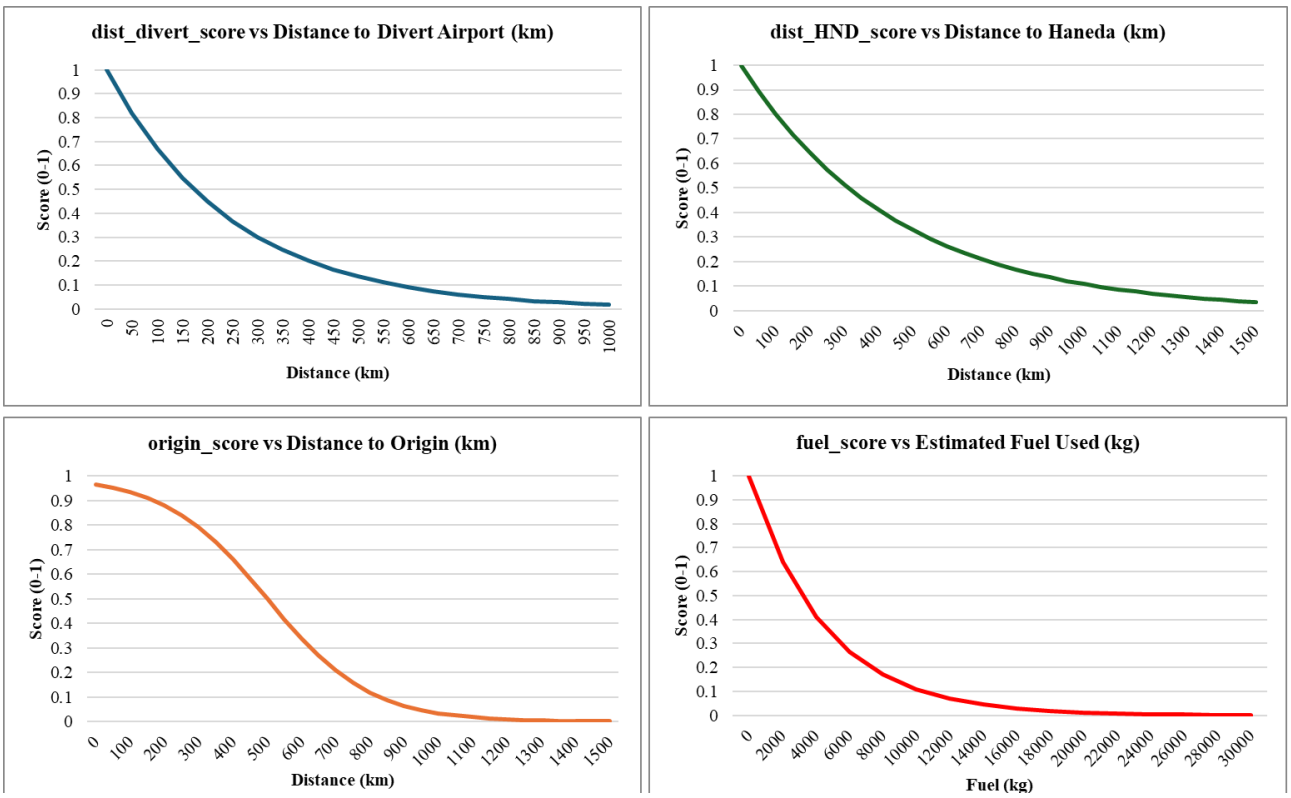


Figure 3-3 Score Decay Curves for Reward Function Components

To illustrate the behavior of each component, Figure 3-3 plots the score as a function of its respective variable:

- dist_divert_score: Rapidly decreases within the first 300 km, indicating a strong preference for nearby diversion airports.
- dist_HND_score: Decreases more gradually (decay constant 450 km), still valuing airports moderately far from Haneda.
- origin_score: Logistic curve sharply penalizes returns to the origin once the distance exceeds ~500 km.
- fuel_score: Exponentially decays with fuel required, with 4500 kg as the decay constant, prioritizing fuel-efficient diversions.

These curves clearly show the intended operational logic — rewarding closer, more fuel-efficient options and penalizing long distances or returns to the origin airport

These four scores are combined via a weighted sum and scaled as shown in eq (1):

$$Reward = (w_1 \cdot \text{dist_divert_score} + w_2 \cdot \text{dist_HND_score} + w_3 \cdot \text{origin_score} + w_4 \cdot \text{fuel_score}) \times 200 \quad (1)$$

The weights w_1, w_2, w_3, w_4 differ depending on the reward perspective.

Fuel-efficiency-centric reward (airline-oriented)

In the airline-oriented reward design, the primary focus is on fuel efficiency and operational feasibility, reflecting the airline's cost and safety priorities. The total reward is calculated as a weighted sum of four components: proximity to the original destination, fuel efficiency, slot congestion, and feasibility-related penalties. The assigned weights are as follows:

$$w_1 = 0.2, w_2 = 0.2, w_3 = 0.2, w_4 = 0.4$$

The primary feasibility criterion in this model is the availability of sufficient fuel for the aircraft to reach the selected diversion airport without refueling. If the estimated fuel required for the diversion exceeds the remaining fuel at the decision point, a strong penalty of −100 is applied to discourage such infeasible choices.

Slot congestion is also penalized to model the limited capacity of each airport. The slot system divides time into 5-minute windows, each with a predefined capacity. If the number of expected arrivals exceeds this capacity, a congestion penalty is applied using eq (2):

$$\text{Penalty} = -10 \times (\text{actual arrivals} - \text{slot capacity} + 1) \quad (2)$$

In the case that no valid slot is available at the selected diversion airport—meaning all time the windows are fully congested—the action is marked as invalid and a fixed penalty of −50 is assigned using eq (3):

$$P_{slot} = \begin{cases} -50, if\ slot_{availabe} = 0 \\ 0, otherwise \end{cases} \tag{3}$$

where slot_available=0 indicates that all 5-minute time windows at the selected diversion airport are already fully occupied.

Additionally, the system accounts for return-to-origin decisions. The departure airport is dynamically added to the action space as a valid alternative. However, to discourage unnecessary returns, the origin_score component reduces the overall reward associated with choosing this option. It is designed to act as a fallback only when no other diversion airport is feasible.

Passenger-oriented reward (env_with_slot_fixed_passenger.py)

To address the post-diversion burden experienced by passengers, a separate reward model is developed in env_with_slot_fixed_passenger.py, emphasizing convenience from the passenger's perspective. The reward function modifies the base structure by adjusting the weight configuration as follows:

$$w_1 = 0.2, w_2 = 0.5, w_3 = 0.2, w_4 = 0.1$$

This model introduces an additional passenger bonus score, which transforms travel inconvenience into a bonus reward. The score is derived from three key metrics that reflect how difficult it is for a diverted passenger to return to Haneda Airport after landing at an alternative airport:

- Total travel time (in minutes)
- Number of transfers (e.g., bus, train)
- Travel cost (in Japanese Yen)

These values were retrieved using Google Maps (Appendix B) and normalized using the maximum observed values to ensure comparability across metrics. The normalization parameters are shown in Table 3-3:

Table 3-3 Normalization Parameters for Passenger Convenience Metrics

Metric	Normalization	Max Value
Time (t)	$t/247$	247 min
Transfers (n)	$n/4$	4 steps
Cost (c)	$c/17480$	¥17,480

Using these, a weighted discomfort score is calculated by eq (4):

$$\text{Discomfort} = 0.4 \cdot \frac{\text{t}}{247} + 0.3 \cdot \frac{\text{n}}{4} + 0.3 \cdot \frac{\text{c}}{17480} \tag{4}$$

Then, the passenger bonus reward is determined by inverting the discomfort score, scaled by a factor of 10 as show in eq (5):

$$\text{PassengerBonus} = (1 - \text{Discomfort}) \times 10 \tag{5}$$

This reward is added to the base reward to promote airport selections that minimize the inconvenience of going to Haneda. As a result, airports such as Narita (NRT) typically receive higher passenger bonuses due to their proximity, lower costs, and fewer required transfers, whereas airports like Kansai (KIX) or Sendai (SDJ) are assigned lower bonuses due to their distance and complexity of ground transportation.

Penalty logic (common to both models)

These penalties are used as constraint-aware shaping terms on top of the base reward (weighted scores × 200). They serve two roles: (i) to mask infeasible actions (hard constraints) and (ii) to discourage congested choices (soft constraints) without destabilizing learning. Magnitudes were calibrated against the base scale so that severe infeasibility dominates the signal (−100), while softer capacity effects remain comparable (−10 per excess arrival).

How penalties integrate with the reward.
Let Base_t denote the weighted score at step *t* (distances/fuel/origin/capacity) and Bonus_t any model-specific add-on (e.g., PassengerBonus = 10·(1 − Discomfort)). The step reward is shown as eq (6)

$$R_t = 200 \cdot Base_t + Bonus_t - 100\{\text{fuel_insufficient}] - 50\{\text{no_slot}\} - 10 \cdot \text{excess_arrivals} \tag{6}$$

- Fuel insufficient (−100). Hard safety constraint. If the remaining fuel cannot cover the modeled requirement to reach the airport under current winds/performance, the action is rejected and a severe penalty (−100) is applied to decisively steer the policy away from unsafe

choices.

- No arrival slot (−50). Operational hard constraint. The action is rejected; a moderate penalty discourages repeated selection of unavailable airports.
- Over capacity (−10 × excess). Soft constraint. The action is allowed but penalized in proportion to the number of arrivals beyond the airport's capacity window, promoting load balancing across alternates.
- Return-to-origin. There is no extra explicit penalty here to avoid double counting. If returning is discouraged in a given experiment, it is reflected implicitly via a lower origin_score inside the base term (and therefore through 200·Base_t). This keeps the modeling choice transparent.

Evaluation order / precedence.
To avoid double penalization when multiple conditions hold simultaneously, checks are performed in the following order:

1. fuel insufficiency → apply −100 and reject the action;
2. no slot available → apply −50 and reject the action;
3. else, compute soft capacity penalty: −10 × excess arrivals (action proceeds).
   (If 1) or 2) is triggered, capacity is not evaluated for that step.)

What "reject" means in the environment.
When an action is rejected (fuel/no-slot), the environment does not transition to that airport; the agent remains in the current decision state (time can advance by one step in the simulator), receives the penalty, and must choose again. This prevents infeasible paths while still providing a strong learning signal.

Table 3-4 Penalty Conditions and Values in the DQN Environment

Condition	Penalty Applied
Remaining fuel < fuel required	-100
No arrival slot available at airport	-50 (action is rejected entirely)
Arrival slot is over capacity	-10 × (number of excess arrivals)
Return to origin (if discouraged)	Implicit penalty via lower origin_score

3.2-4 Simulation procedure and deployment

To evaluate the effectiveness of the trained DQN agents under realistic flight diversion scenarios, the author designed a simulation pipeline that mimics the actual operational decision process at the time of an airport disruption. This section describes the simulation logic, data preparation, and result output methodology for both fuel-efficiency and passenger-convenience models.

Simulation objective

The goal of the simulation is to test the trained policy by presenting it with a set of actual aircraft states and comparing its selected diversion airport with various evaluation metrics such as fuel feasibility, proximity, and post-diversion convenience.

Simulations are executed using two separate DQN models:

- Fuel-focused model: dqn_model_eta_based_fuel.py
- Passenger-focused model: dqn_model_eta_based_passenger.py

Both models are applied to the same input dataset.

Input data

The simulation process utilizes three primary datasets. First, the file real_flight_states_with_position_final.csv contains the preprocessed aircraft states recorded at the moment of disruption (08:47 UTC). Each record in this file includes the aircraft's latitude and longitude, the great-circle distance to Haneda Airport, altitude (in meters), ground speed (in meters per second), remaining fuel (in kilograms), and the estimated fuel required to return to the departure airport (fuel_to_origin).

The second input file, fuel_remainging_results.csv, provides precomputed values for each flight-airport pair. These values include the estimated fuel consumption required to reach a specific diversion airport and the aircraft’s remaining fuel at the time of the diversion decision. This file ensures that fuel feasibility is considered during both reward calculation and action validation.

In addition to numerical states, each flight record includes metadata such as the flight ID, aircraft type, and departure airport code, all of which are used to dynamically configure the simulation environment and track the results per aircraft.

Simulation procedure

This section describes the simulation process for evaluating diversion decisions using Deep Q-Networks (DQN), as illustrated in Figure 3-4.
The process is implemented in two primary scripts: simulate_dqn_all_fuel.py for the fuel-efficiency-focused model and simulate_dqn_all_passenger.py for the passenger-convenience-oriented model. Both scripts follow a common structure but differ in their environment logic and reward interpretation.

The simulation begins by loading a pre-trained DQN model built using PyTorch[17]. The model is transferred to the CPU environment to ensure broad compatibility and ease of batch simulation.

For each aircraft record in the dataset, a custom RerouteEnv environment is instantiated. This environment is initialized with the following parameters:

- Geographic coordinates for five predefined diversion airports (Narita, Nagoya, Kansai, Osaka Itami, and Sendai), along with the aircraft's dynamic origin airport.
- Slot availability and capacity constraints (e.g., a maximum of 2 arrivals per 5-minute slot).
- The original destination airport, fixed as Haneda (HND).

Next, a state vector representing the aircraft's situation at the decision point—typically including latitude, longitude, altitude, speed, remaining fuel, and aircraft type—is injected into the environment. This vector is accompanied by metadata specifying the aircraft’s origin and performance model.

Action selection proceeds by computing Q-values for all available diversion options. Each airport's feasibility is assessed:

- If fuel is insufficient to reach the airport, the resulting reward is penalized (e.g., −100).
- If no valid slot exists due to congestion, the action is rejected with a fixed reward of −50.

The selected action is passed into the environment via env.step(action), which returns several outputs:

- The total reward for the action
- A validity flag indicating whether the action was acceptable
- The selected airport code
- Individual reward components including dist_divert_score, dist_HND_score, origin_score, and fuel_eff_score
- Applied penalties related to slot congestion or infeasibility

Each result is appended to a Pandas DataFrame and ultimately saved as a CSV file. The file name reflects the evaluation perspective:

- simulation_results_reroute_finalfuel.csv for the fuel-efficiency simulation
- simulation_results_reroute_finalpassenger.csv for the passenger-oriented simulation

Each output row contains detailed evaluation metadata, structured as follows:

> {"index", "lat", "lon", "chosen_airport", "reward", "valid", "dist_divert_score", "dist_HND_score", "origin_score", "fuel_eff_score", "penalty", "origin_airport", "is_origin_return}

This simulation framework enables a comprehensive evaluation of the DQN model's decision-making under varying reward criteria, supporting later comparative analyses of airline- and passenger-oriented policies.

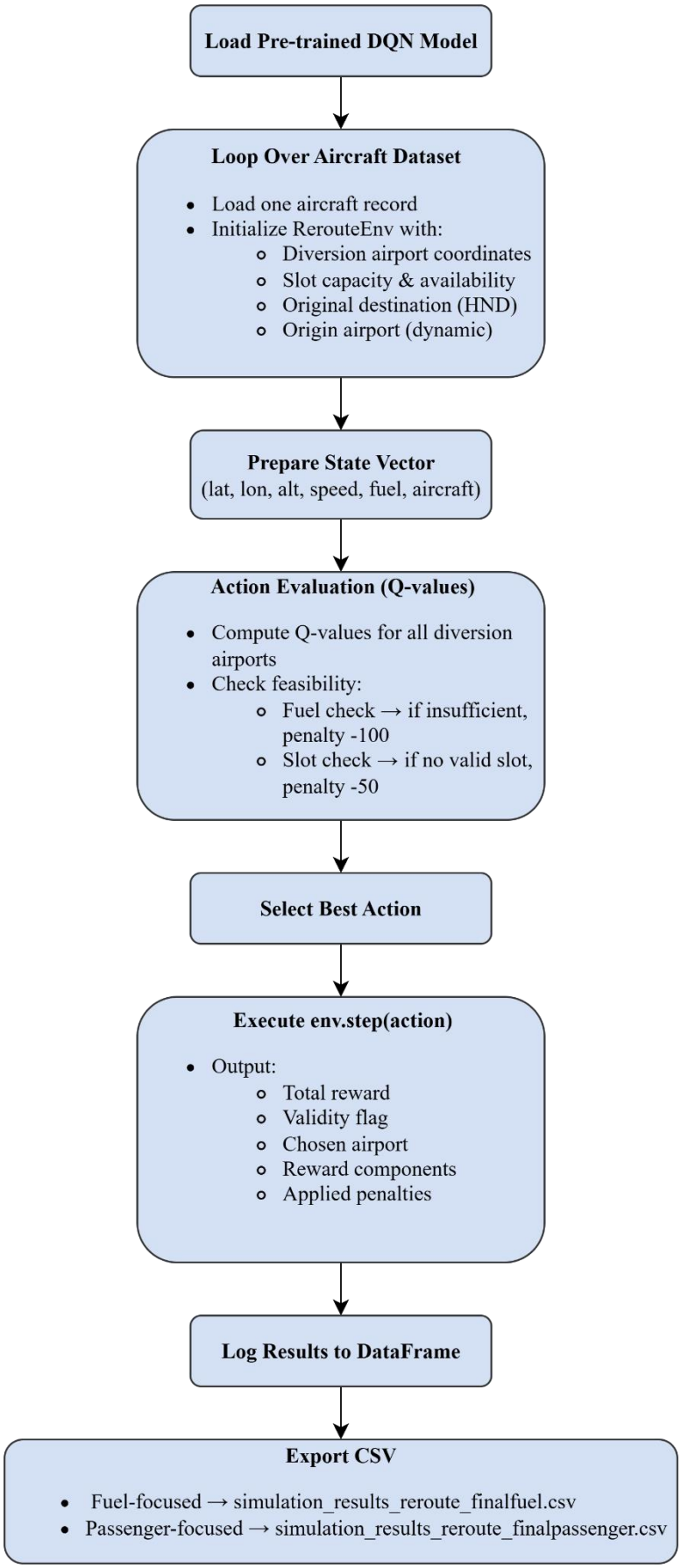


Figure 3-4 Simulation flow for DQN-based Diversion Decision-making

Dynamic action handling

The departure airport is dynamically added to the candidate list if not already present. This allows DQN to consider returning to the origin as a legitimate fallback option under fuel or slot constraints.

- The number of output Q-values dynamically adjusts using:

$$\text{Q_values}[:, : \text{len}(\text{env.airport_coords})]$$

- Output dimension is capped by MAX_OUTPUT_DIM = 10 (to accommodate dynamic extension).

Deployment scalability

The system is designed to process any number of aircraft records by iterating row-wise through the input dataset. Each flight simulation runs independently and results can be analyzed individually or in aggregate.

No parallelization was applied in the current implementation, but the structure is compatible with vectorized batch execution or multiprocessing.

3.2-5 Input data required for DQN system

The proposed DQN-based aircraft diversion optimization system relies on accurate and comprehensive input data to simulate realistic decision-making scenarios. In particular, the system requires three categories of input:

1. The state of each aircraft at the time of disruption,
2. The estimated remaining fuel onboard,
3. The projected fuel required to divert to each candidate airport.

These inputs are generated through a combination of real-time flight tracking data (via Flightradar24), a fuel estimation model based on BADA aircraft performance data, and a flight trajectory optimization tool that simulates fuel-optimal diverting.

To clarify the relationship between these data components, Figure 3-5 illustrates a representative example flight from origin to destination. The trajectory segments and decision point are explicitly annotated to indicate how each fuel-related quantity is computed:

- The yellow segment shows the actual flight path from the origin airport to the point at disruption time (08:47 UTC), representing the fuel used.

- An aircraft icon indicates the aircraft's position at 08:47 UTC. This location is used to define the initial state vector provided as input to the DQN.
- The purple line represents the diversion path simulated after disruption, used to estimate the fuel required to divert.
- The blue line is the originally planned trajectory before the disruption event.

This spatial decomposition is essential for calculating fuel feasibility and simulating realistic outcomes during training and inference.

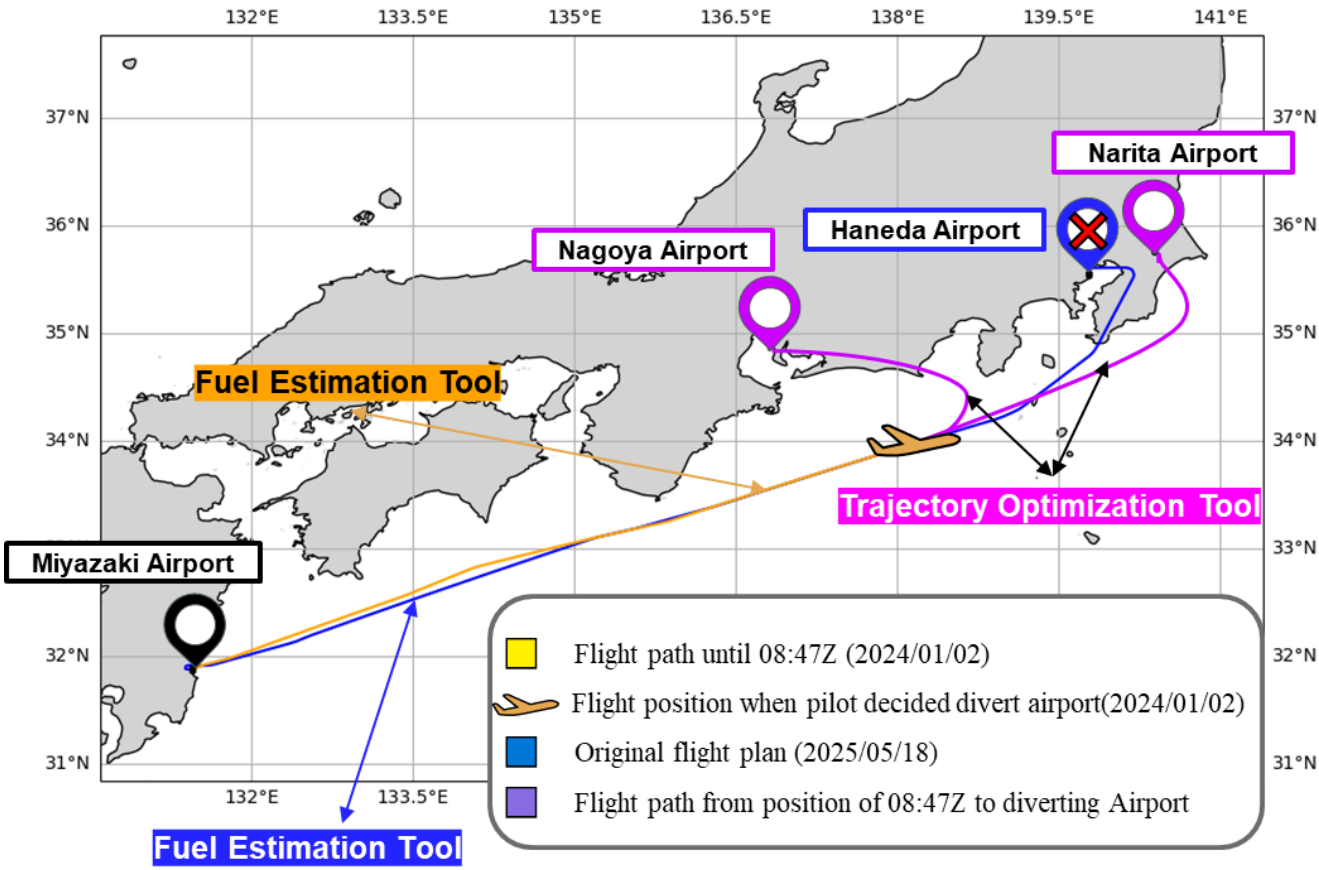


Figure 3-5 Decomposition of aircraft trajectory for fuel estimation at disruption point (08:47 UTC)

3.3 Flight data collection – Flightradar24

In order to construct a realistic learning and simulation environment, trajectory data of aircraft scheduled to arrive at Haneda Airport between 08:47 and 12:30 UTC on January 2, 2024, were collected based on the Scheduled Time of Arrival (STA). These data were obtained through Flightradar24, a real-time aircraft tracking platform based on Automatic Dependent Surveillance–Broadcast (ADS-B) technology. Each record includes timestamp, UTC time, callsign, position (latitude/longitude), altitude, speed, and heading. The data were downloaded in CSV format.

An example of the raw trajectory data is shown in Table 3-5. Table 3-6 describes the columns used.

Table 3-5 Example of ADS-B flight data (6J62)

Timestamp	UTC	Callsign	Position [Lat, Lon]	Altitude (ft)	Speed (knots)	Direction
1704182557	2024-01-02T 08:02:37Z	SNJ62	31.877415,131.450974	0	145	84
1704182564	2024-01-02T 08:02:44Z	SNJ62	31.877855,131.456909	0	160	85
1704182573	2024-01-02T 08:02:53Z	SNJ62	31.878508,131.464752	325	158	84
1704182582	2024-01-02T 08:03:02Z	SNJ62	31.879578,131.472626	675	164	77
1704182586	2024-01-02T 08:03:06Z	SNJ62	31.880276,131.475403	800	166	73
1704182589	2024-01-02T 08:03:09Z	SNJ62	31.881021,131.478104	950	166	72
1704182592	2024-01-02T 08:03:12Z	SNJ62	31.881672,131.48056	1100	166	72
1704182595	2024-01-02T 08:03:15Z	SNJ62	31.882416,131.483109	1250	164	71
1704182598	2024-01-02T 08:03:18Z	SNJ62	31.883148,131.48558	1375	165	70
1704182601	2024-01-02T 08:03:21Z	SNJ62	31.884109,131.488434	1525	165	69

Table 3-6 Unit and description of ADS-B flight data

Column Name	Unit	Description
Timestamp	Epoch time (sec)	Time of data recording in UNIX format, measured in seconds since 1970-01-01T00:00:00 UTC.
UTC	ISO 8601 format	Coordinated Universal Time in standard ISO 8601 format (e.g., 2024-01-02T08:02:37Z)
Callsign	-	Unique identifier assigned to the aircraft
Position	Degrees	Geographic coordinates indicate aircraft position in decimal degrees.
Altitude	Feet (ft)	Altitude of the aircraft above sea level at the given time.
Speed	Knots	Ground speed of the aircraft; 1 knot ≈ 1.852 km/h
Direction	Degrees	Heading of the aircraft in degrees clockwise from true north (0–360°).

Out of the initially collected 78 flight trajectories, 19 were excluded through filtering due to missing data within ±30 minutes of the accident time (08:47 UTC) or due to incomplete trajectories (e.g.,

large temporal gaps). As a result, 59 valid flights remained for analysis. These filtered flight data were subsequently used as input for the fuel estimation tool, the flight trajectory optimization tool, and the DQN-based diverting simulation.

3.4 Fuel remaining data - fuel estimation tool

In this study, a fuel estimation tool developed by Associate Professor Harada's laboratory [18] was employed to estimate the cumulative fuel consumption of each aircraft using actual ADS-B trajectory data. This module plays a crucial role in evaluating the operational feasibility and fuel efficiency of AI-based diverting decisions by quantitatively determining the amount of fuel consumed by each aircraft up to a specific point in time (e.g., the moment of disruption).

The estimation system integrates three primary data sources:

- ADS-B trajectory data (latitude, longitude, altitude, and time) obtained from Flightradar24,
- Reanalysis meteorological data from ERA5 provided by the Copernicus Climate Data Store[1] https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels
- Aircraft performance models derived from Eurocontrol's BADA 3.15 Operational Performance Files (OPF).

3.4-1 Time interpolation and preprocessing

The raw ADS-B trajectory records, which are sampled at irregular time intervals, are linearly interpolated to generate uniformly spaced data at 10-second intervals. This step ensures temporal consistency for aircraft performance modeling and atmospheric data matching. The following variables are resampled:

- Time (UNIX timestamp in UTC),
- Latitude and Longitude,
- Pressure Altitude (ft).

The interpolation is mathematically expressed as:

$$(t_i') = interp1\big(t, f(t)\big) \tag{7}$$

where $f(t)$ represents a flight variable (e.g., altitude), t_i are the original timestamps, and t_i' are the new 10-second-spaced timestamps.

3.4-2 Meteorological matching and interpolation

Each interpolated flight point is mapped to the ERA5 dataset, which is provided on a three-dimensional grid (latitude, longitude, altitude) and updated every 6 hours (00Z, 06Z, 12Z, 18Z).

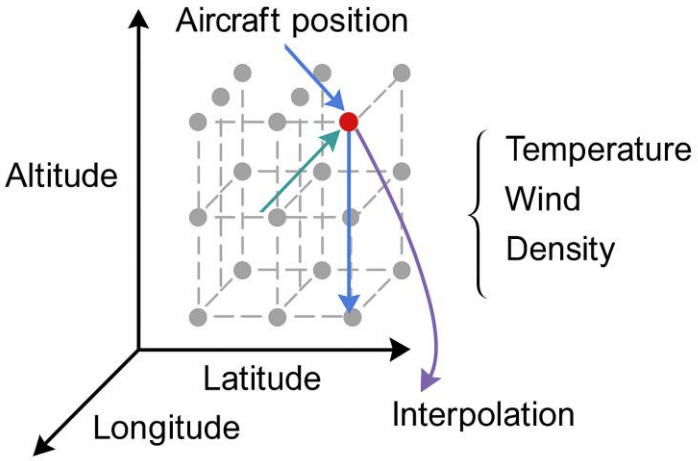


Figure 3-6 Conceptual Mapping of Aircraft Position onto 3D Meteorological Grid

Figure 3-6 illustrates the spatial relationship between an interpolated aircraft position and the surrounding ERA5 meteorological grid nodes. Because aircraft locations rarely coincide with grid nodes, bilinear spatial and vertical (altitude-wise) interpolation is applied.

If the flight timestamp lies between two reanalysis slices (e.g., 06Z and 12Z), linear interpolation is performed in the temporal dimension to estimate the meteorological variables at that time.:

- T: Temperature (°C)
- ρ: Air density (kg/m³)
- u, v: Wind vector components (m/s)

These interpolated meteorological inputs are then used to compute true air speed, vertical profile, aerodynamic drag, engine thrust, and fuel flow at each point. Finally, the cumulative fuel burn is derived by time integration using BADA-based Specific Fuel Consumption (SFC) coefficients.

3.4-3 Velocity estimation

Based on the interpolated trajectory and meteorological data, several key velocity parameters are computed to characterize the aircraft's flight profile:

- Ground Speed (V_{GS}): Calculated using the haversine distance between consecutive GPS points and corresponding timestamps.

- True Airspeed (V_{TAS}): Derived by correcting the ground speed with interpolated wind vectors
- Calibrated Airspeed (V_{CAS}): Estimated from the true airspeed using a standard TAS–CAS mat file.

Once the velocity parameters are established, the following aerodynamic forces and coefficients are calculated using the interpolated atmospheric data (pressure altitude, air density) and the aircraft's performance model:

$$C_L = \frac{L}{0.5\rho V^2 S}, \qquad C_D = C_{D0} + C_{D2}C_L^2 \tag{8}$$

$$D = 0.5\rho V^2 S C_D \tag{9}$$

Where:

- C_L: Lift coefficient
- C_D: Drag coefficient
- ρ: Air density at flight altitude
- V: True airspeed (V_{TAS})
- S: Wing [m^2]
- C_{D0}, C_{D2}: Empirical drag coefficients provided by the BADA OPF model
- L: Required lift (typically equal to aircraft weight in steady-level flight)
- D: Drag force experienced by the aircraft [N]

3.4-4 Thrust and fuel flow estimation

To determine the instantaneous fuel consumption, the required engine thrust is first estimated at each time step using the following force-balance eq (10):

$$T = D + m\frac{dV}{dt} + mg\sin(\gamma) \tag{10}$$

Where:

- T: Required thrust
- D: Aerodynamic drag (from Equation 3)
- m: Aircraft mass (assumed constant during each segment) [kg]

- $\frac{dV}{dt}$: Aircraft acceleration along the flight path [m/s^2]
- g: Gravitational acceleration [9.80665 m/s^2]
- γ: Flight path angle derived from altitude gradient [rad]

The calculated thrust is capped at the altitude-specific maximum thrust defined in the BADA 3.15 model to ensure physical feasibility.

Subsequently, the fuel flow is estimated using a proportional relationship between thrust and fuel consumption:

$$\text{Fuel Flow} = \eta \cdot T \tag{11}$$

Where:

- η: Specific Fuel Consumption (SFC) coefficient obtained from BADA OPF for the given flight phase and altitude $[\text{kg}/(\text{N} \cdot \text{s})]$
- T: Thrust (as calculated above)

In cruise conditions or when vertical speed is negligible, a pre-defined cruise fuel flow model is used instead, consistent with the BADA methodology.

3.4-5 Cumulative fuel burn

Each instantaneous fuel flow value (estimated at 10-second intervals) is multiplied by the time interval (Δt = 10 s) to compute the amount of fuel consumed during that interval. The total fuel consumption TFC is then obtained by summing overall time steps as follows:

$$\text{TFC} = \sum_{i=1}^{n} \text{FuelFlow}_i \cdot \Delta t \tag{12}$$

The aircraft's mass is simultaneously updated as:

$$m_{i+1} = m_i - \text{FuelFlow}_i \cdot \Delta t \tag{13}$$

3.4-6 Output results

The Fuel Estimation Tool computes time-resolved estimates of aircraft mass and fuel flow based on ADS-B trajectory and interpolated weather. The key outputs are:

Table 3-7 Key Result of Fuel Estimation Tool

Output Variable	Unit	Description
result_est(i).est_mass	kg	Time series of aircraft gross weight, updated as fuel is burned.
result_est(i).est_FF	kg/s	Fuel flow rate at each time step along the flight path.
TFD(i).fuel_used	kg	Total fuel burned until the accident time (e.g., 08:47 UTC).
TFD(i).trip_distance	km	Distance flown until that point.
TFD(i).trip_time	S	Duration from departure to 08:47 UTC.

A complete list of output variables and data structures used in the fuel estimation module is provided in Appendix C.

To evaluate fuel feasibility in real-time diverting scenarios, the Fuel Estimation Tool is used to calculate two types of fuel consumption from ADS-B trajectory data.

First, for 59 selected aircraft operating on January 2, 2024, the tool estimates the amount of fuel consumed from takeoff to the time of disruption (08:47 UTC). These flights were filtered from Flightradar24 ADS-B records based on their trajectory toward Haneda Airport. Since ADS-B data are sampled irregularly and may not contain an exact timestamp of 08:47:00 UTC, the record closest to 08:47UTC (e.g., 08:47:05UTC) is selected for each flight.

Second, to estimate the trip fuel, the tool uses the same aircraft operating on a different day—in this study, flights from February 2nd, 2024—and calculates fuel consumption from takeoff to landing. These reference flights are assumed to represent a standard flight profile under normal operations.

By comparing the fuel used up to 08:47 UTC with the full-trip fuel, the remaining fuel at the time of disruption can be inferred. This remaining fuel is then used in simulations to assess the feasibility of diversion to alternate airports.

3.5 Fuel used data – flight trajectory optimizer

To optimize the post-accident diverting trajectories of aircraft from the perspective of fuel efficiency, this study utilizes flight Trajectory Optimization Tool. This tool takes the aircraft's current position, speed, altitude, remaining fuel, and surrounding meteorological conditions as input and computes a physically feasible trajectory to a designated diversion airport that minimizes total fuel consumption. The optimized trajectory provides a quantitative basis for verifying whether AI-generated diverting decisions are viable under actual atmospheric and aerodynamic constraints. It also enables detailed comparison with historical trajectories and estimated fuel burn.

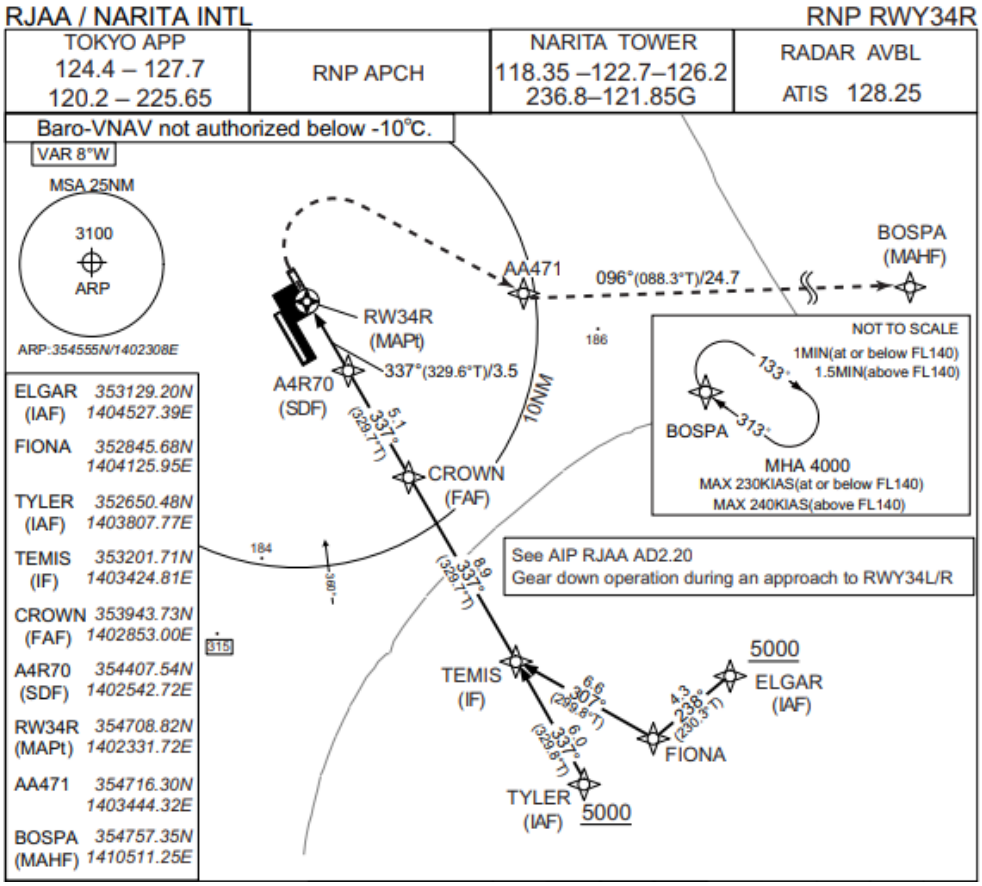
3.5-1 Optimization target and scenario setup

The optimization targets comprise 59 commercial aircraft that were en route to Tokyo Haneda Airport (RJTT) on January 2, 2024, at the time of the accident (08:47 UTC). For each aircraft, a set of five alternate diversion airports was predefined, including Narita (NRT), Nagoya (NGO), Osaka Itami (ITM), Kansai (KIX), and Sendai (SDJ). Accordingly, a total of 295 diverting scenarios (59 aircraft × 5 candidate airports) were constructed for optimization.

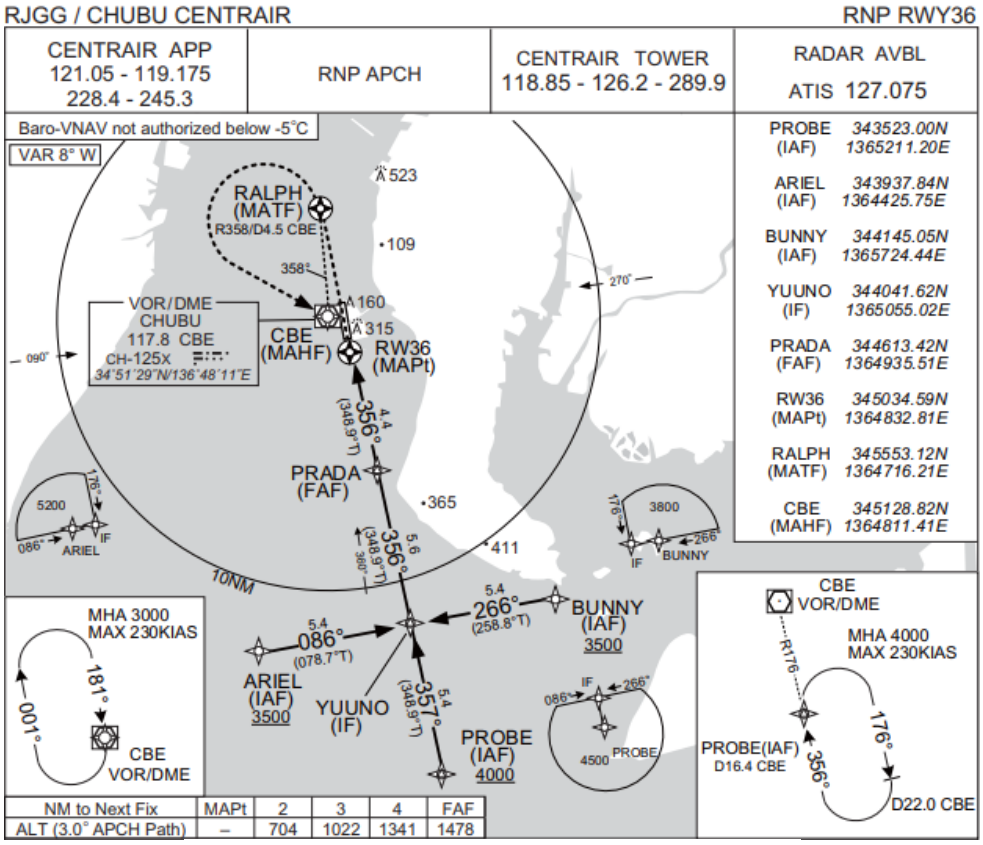
For each diverting scenario, the tool receives the following initial and boundary conditions:

- Initial position: The aircraft's location at 08:47 UTC on January 2, 2024, as derived from ADS-B data.
- Destination point: The Initial Fix (IF) waypoint of the standard RNAV arrival procedure (RNP APCH) for the corresponding runway, derived from official AIP Japan RNAV approach charts.[19]
- Initial state parameters: Estimated from flight data, including V_{cas}, altitude, and gross weight at the time of accidenty.
- Final Constraints: The optimized trajectory must terminate at the RNAV IF waypoint at a fixed altitude, following the procedural minimum specified in the AIP charts. The required airspeed at this final point is derived from the aircraft's performance characteristics in the BADA 3.15 Performance Table File (PTF) at the corresponding altitude.
- Feasibility Constraints: The optimization ensures that the entire trajectory remains within the feasible flight envelope, including thrust limitations, maximum climb/descent rates, and fuel availability, as defined by the BADA 3.15 performance model.

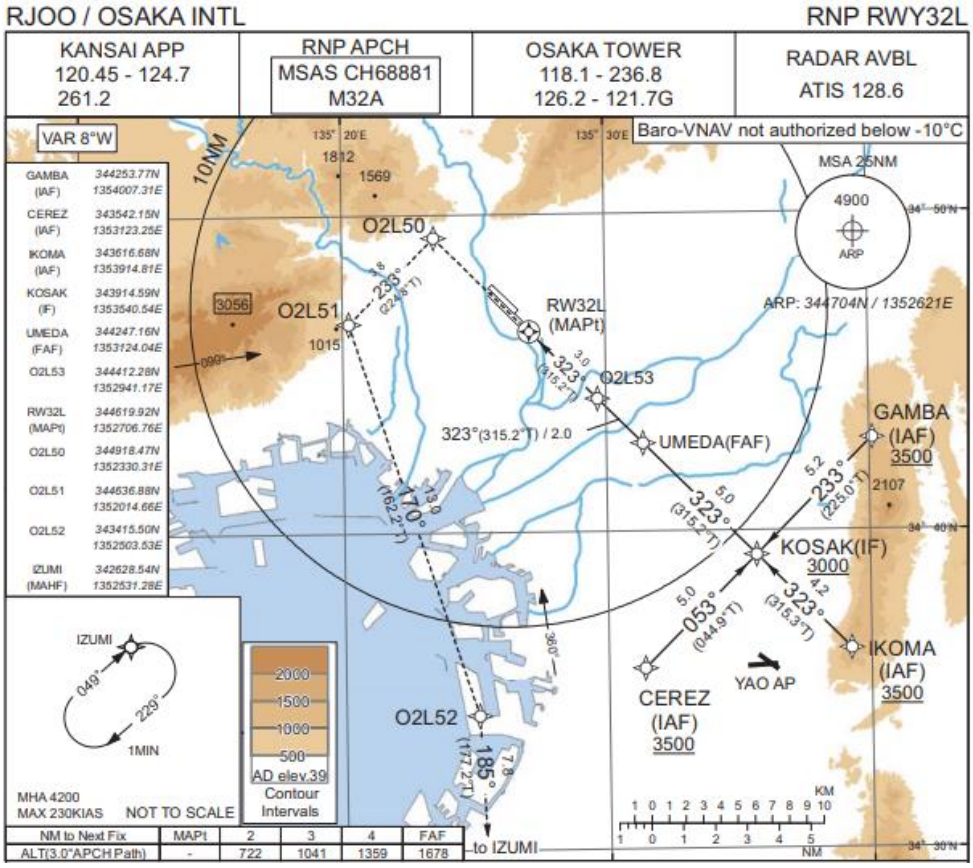
Each scenario is solved independently by the optimization module, allowing for direct comparison of fuel-optimal diverting options across various candidate airports. The resulting trajectories also serve as ground truth for evaluating whether AI-based decisions (e.g., from the DQN model) are physically achievable given the aircraft's post-accident condition.



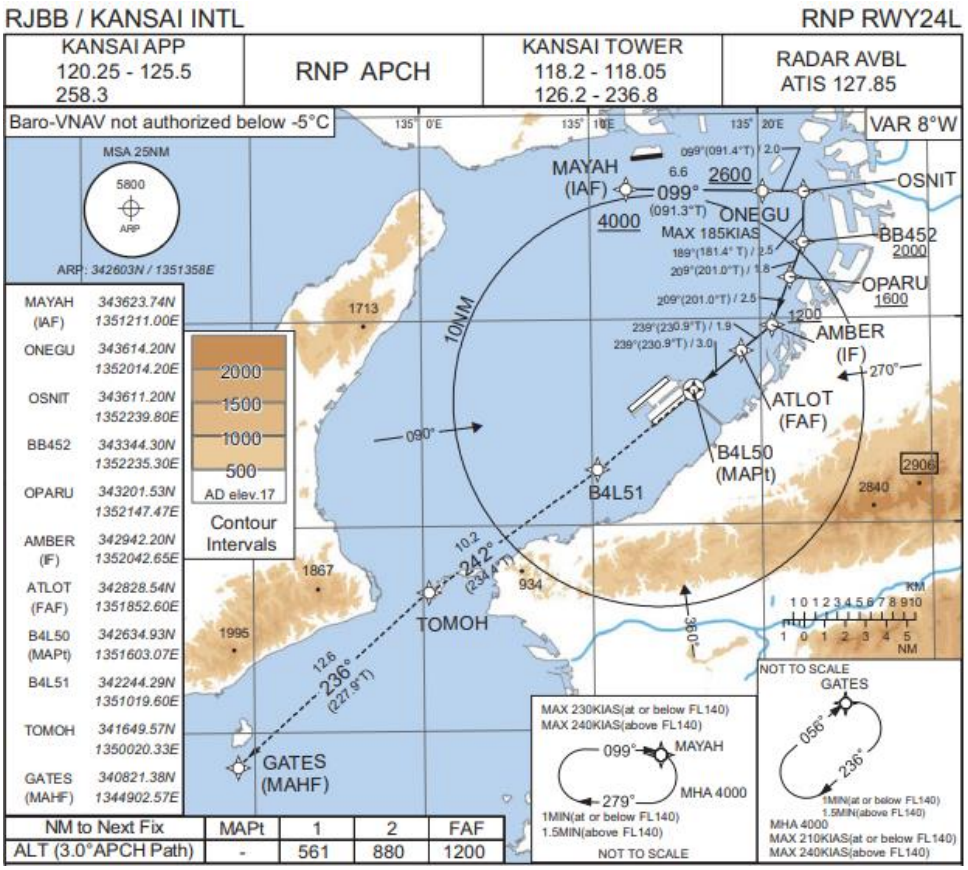
(a) RNAV Arrival Charts for Narita Airport



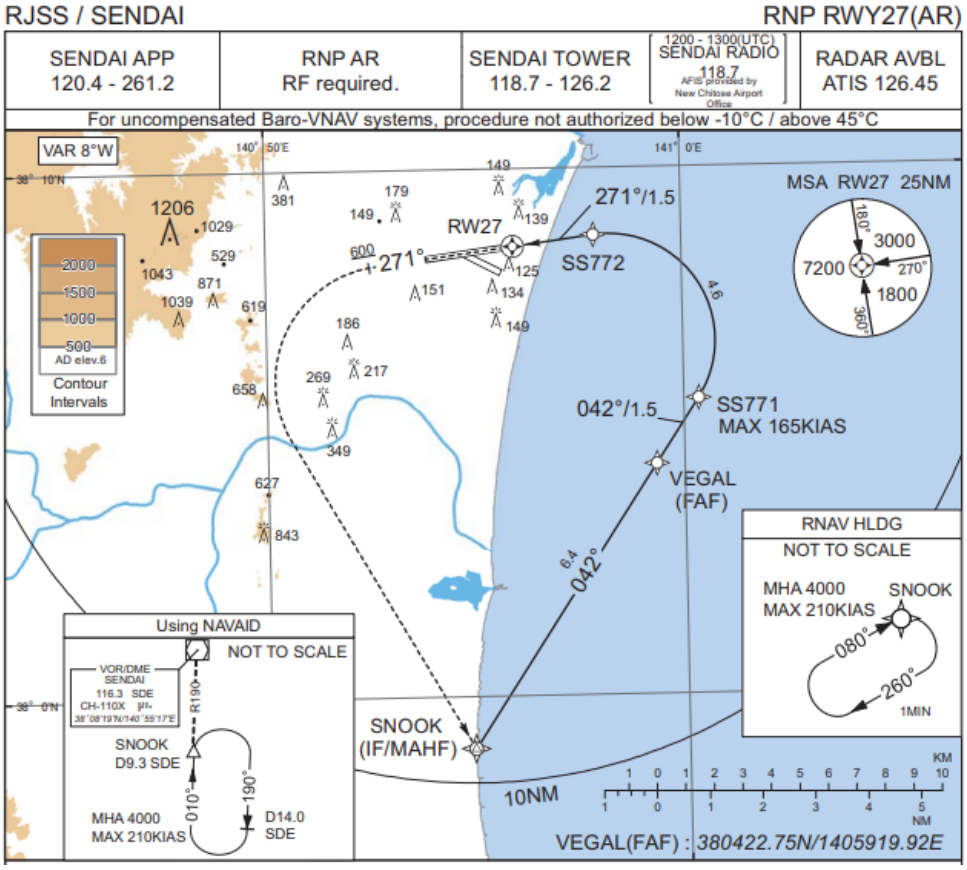
(b) RNAV Arrival Charts for Nagoya Airport



(c) RNAV Arrival Charts for Itami Airport



(d) RNAV Arrival Charts for Kansai Airport



(e) RNAV Arrival Charts for Sendai Airport

Figure 3-7 RNAV Arrival Charts for NRT, NGO, ITM, KIX, and SDJ (Source: AIS Japan)

These IF points serve as the final destinations for the trajectory optimization process, enabling seamless transition into the published RNAV approach segment. Refer to Figure 3-7 for visualizations extracted from Japan AIP RNAV approach charts.

3.5-2 Input data structure

The flight Trajectory Optimization Tool utilizes a structured set of input files that provide environmental conditions, aircraft characteristics, and trajectory parameters for each diverting case.

Table 3-8 Flight Trajectory Optimizer Input Data

Category	File Name	Description
Meteorological Data	metdata_GSM2_20240102.mat	3D reanalysis weather data (T, ρ, wind) from JMA GSM2 model.
Boundary Conditions	BC_of_AC_20240102_Alltypes.data	Initial/final positions speed, mass per diverting scenario.
Aircraft Type List	ACtype_list.mat	ICAO type codes (e.g., B738, A320) for each aircraft case.
Performance Profiles	BADA_OPF_ACtype.mat	BADA 3.15 coefficients (drag, thrust, SFC) per aircraft type.
Grid Resolution (nx)	nx_values.mat	Number of trajectory grid points based on flight distance (~ 18km spacing).
Optimization Module	main_MSDP_Function_mex_nxXX.mexw64	Precompiled MEX optimizers customized for each nx value.

Each of these input components plays a specific role in ensuring that the optimization process reflects realistic aircraft behavior and environmental constraints. The modular design also supports batch processing of hundreds of diverting scenarios in parallel.

3.5-3 Trajectory grid initialization

To facilitate trajectory optimization, the flight path from the aircraft's current position to the selected diversion airport is discretized into uniformly spaced segments. This discretization defines the optimization variables and offers a physically meaningful initial guess, which improves both convergence speed and solution stability.

Grid construction

For each diverting scenario, the great-circle distance between the accident point and the designated alternate airport is divided into intervals of approximately 18 km. The total number of grid points n_x is scenario-specific and stored in the nx_values.mat file.

Initial trajectory setup

Each grid point x_k is initialized by linearly interpolating between the start and end positions in three dimensions—latitude, longitude, and altitude. This produces a smooth initial trajectory x(t), which acts as the starting guess for the optimizer:

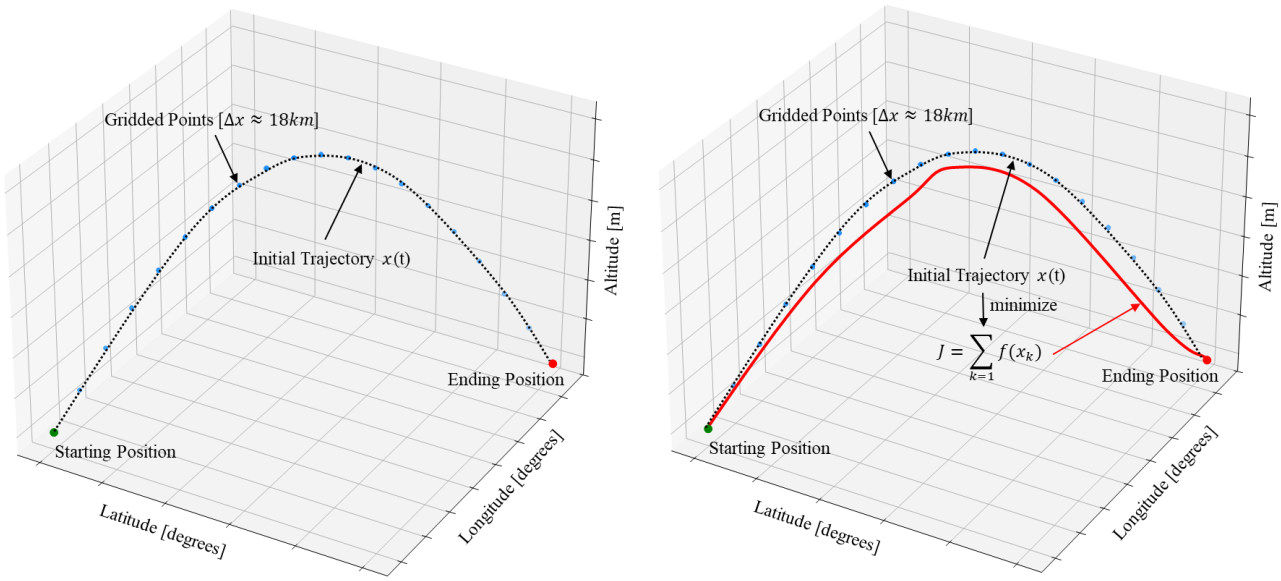
$$x_k = \left(1 - \frac{k}{n_x}\right) x_{start} + \frac{k}{n_x} x_{end}, \quad k = 0,1, \dots, n_x \tag{14}$$

Optimization objective

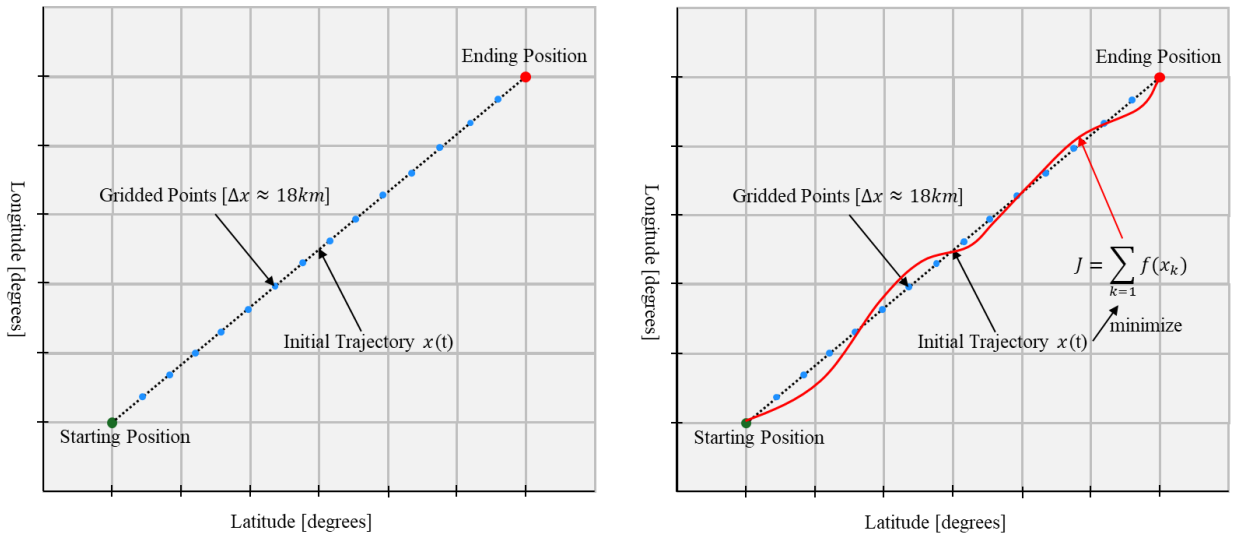
The goal is to minimize the total fuel consumption along the trajectory. The objective function J is defined as the sum of cost values (e.g., fuel flow or drag penalties) at each trajectory segment:

$$J = \sum_{k=1}^{n_x} f(x_k) \tag{15}$$

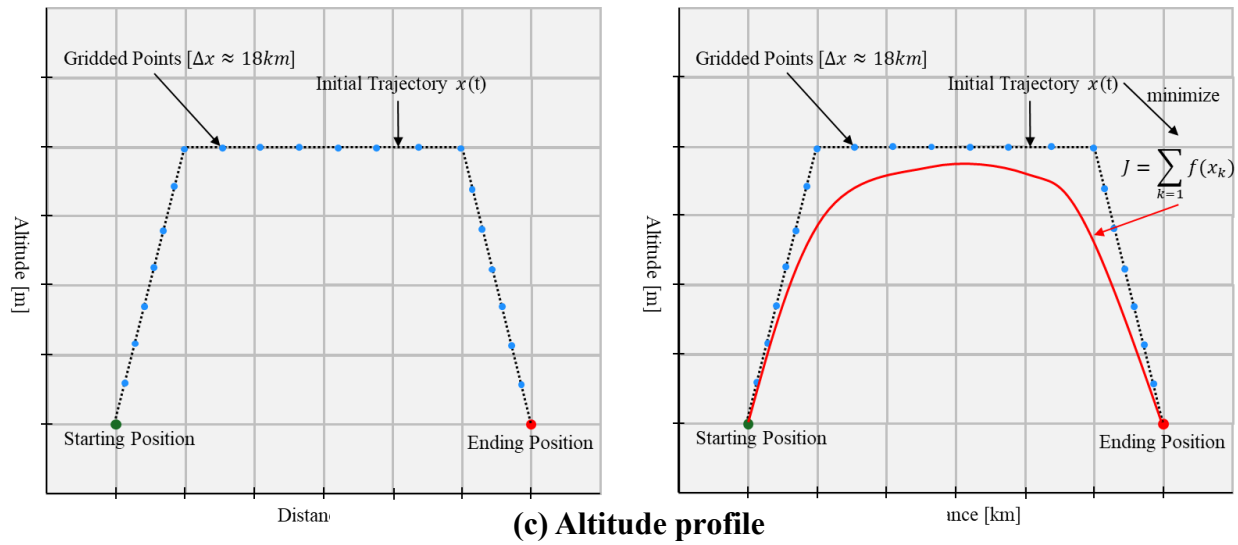
Where $f(x_k)$ denotes the fuel or aerodynamic cost at the k-th grid point, based on BADA performance parameters and interpolated weather conditions.



(a) 3D discretization



(b) Plan view (latitude-longitude)



(c) Altitude profile

Figure 3-8 Flight Trajectory Optimization: 3D, plan view, and vertical profile.

Figures 3-8(a to c) illustrate the discretization and the resulting initial (black dotted) and optimized (red) trajectories in 3-D, plan view (latitude–longitude), and distance–altitude coordinates

3.5-4 Meteorological interpolation

To accurately estimate fuel consumption and aerodynamic forces along the diverting trajectory, meteorological conditions must be interpolated at each discretized point. This process uses reanalysis weather data to extract local atmospheric conditions across time, space, and altitude.

Weather data source

The tool uses the GSM2 dataset provided by the Japan Meteorological Agency (JMA)[20]. This data contains gridded weather variables on a 3D mesh (latitude, longitude, and altitude), with temporal resolution at 6-hour intervals (00Z, 06Z, 12Z, 18Z).

Interpolation procedure

For each trajectory point x_k, the following atmospheric variables are interpolated:

- Temperature $T(x_k)$
- Air density $\rho(x_k)$
- Wind components $u(x_k)$, $v(x_k)$

The interpolation consists of:

1. Horizontal interpolation (bilinear over lat/lon grid)
2. Vertical interpolation (linear over altitude levels)
3. Temporal interpolation (if x_k's timestamp lies between two forecast blocks)

This produces precise local weather values at each grid point, which are fed into aerodynamic and fuel models.

Two-step optimization structure

To improve convergence and prevent instability from abrupt wind shifts, a two-phase optimization approach is employed:

1. No-wind optimization:
    - Meteorological terms (wind, pressure gradients) are ignored

- Produces a physically valid, smooth baseline trajectory

2. Wind-aware optimization:

- Full interpolation is applied to all x_k points
- Wind vectors u, v are incorporated into aircraft dynamics and fuel modeling
- Refines the trajectory to minimize fuel under realistic atmospheric conditions

3.5-5 Aerodynamic computation

At each point along the trajectory, aerodynamic forces and fuel consumption are computed using the interpolated meteorological data and aircraft-specific performance parameters. The calculations follow standard aircraft physics based on the EUROCONTROL BADA 3.15 performance model.[10]

Force and fuel estimation at each grid point

For each trajectory point x_k, the following quantities are calculated:

Table 3-9 Key aerodynamic and fuel-related variables used in trajectory optimization

Variable	Description
V_{TAS}	True airspeed (m/s), derived from interpolated ground speed and wind
$\rho(x_k)$	Air density at x_k, interpolated from GSM2
$D(x_k)$	Drag force, calculated using: $D = \frac{1}{2}\rho V_{TAS}^2 S(C_{D0} + C_{D2}C_L^2)$
$T(x_k)$	Required thrust, accounting for drag and climb/descent force components
$\mathrm{SFC}(x_k)$	Specific Fuel Consumption at given altitude and thrust
$\dot{m}_f(x_k)$	Fuel flow rate: $\dot{m}_f = T \cdot SFC$

Aircraft-specific constants such as C_{D0}, C_{D2}, S (wing area), and SFC tables are extracted from the BADA OPF files pre-loaded in BADA_OPF_ACtype.mat.

Mass update and step-wise integration

Fuel consumption causes continuous mass reduction throughout the trajectory. A recursive update is applied over the grid using the computed fuel flow rate:

$$m_{k+1} = m_k - \dot{m}_f(x_k) \cdot \Delta t \tag{16}$$

- m_k: aircraft mass at segment k
- Δt: time interval between x_k and x_{k+1} (typically 10 seconds)
- Initial mass m_0: derived from fuel estimation results at 08:47 UTC

The aircraft's lift and thrust calculations are adjusted at each step using the updated mass value.

Flight phase-specific modeling

The BADA model separates the flight into three segments:

Table 3-10 Flight phase classification and corresponding BADA fuel behavior

Phase	Key conditions	BADA Behavior
Climb	$\frac{dh}{dt} > 0$	Higher SFC, increasing thrust
Cruise	$\frac{dh}{dt} \approx 0$ and stable V	Efficient SFC, constant altitude
Descent	$\frac{dh}{dt} < 0$	Reduced thrust, low SFC

The optimizer selects feasible climb/descent rates within the aircraft's physical limits and applies corresponding aerodynamic coefficients.

This aerodynamic module ensures that:

- Each trajectory point satisfies physical flight constraints (e.g., thrust < max thrust)
- Fuel flow is accurately modeled using real-time atmospheric and aircraft performance data
- Aircraft mass is updated recursively to reflect fuel burn

Combined with meteorological interpolation and grid-based optimization, this forms the physical backbone of the trajectory evaluation system.

3.5-6 Numerical optimization

Once the initial trajectory is constructed and all meteorological and aerodynamic parameters are defined, the optimization process begins. The objective is to minimize the total fuel consumption over the trajectory, subject to physical and operational constraints.

The optimization algorithm treats the latitude, longitude, and altitude of each internal trajectory point (excluding the fixed start and end) as decision variables. These variables are adjusted iteratively to minimize the total cost function, which includes fuel flow penalties, aerodynamic drag, and climb/descent costs. The cost function is defined in Equation (15).

Optimization algorithm and tool architecture

The optimization routine is implemented in MATLAB using precompiled MEX functions named main_MSDP_Function_mex_nx, where xx corresponds to the number of trajectory grid points (nx) for each scenario.

These functions employ a Sequential Quadratic Programming (SQP) solver internally, offering robust handling of nonlinear constraints and efficient convergence for medium-scale trajectory problems.

Each MEX function includes:

- Objective gradient computation
- Jacobian and Hessian matrix estimation
- Constraint handling using interior-point method or SQP

To minimize computational cost, the tool selects a precompiled MEX solver matching the scenario's nx value, using an index from nx_values.mat.

Constraints

The trajectory optimization process is subject to several physical and procedural constraints to ensure that the resulting path is not only fuel-efficient but also operationally feasible.

- Initial conditions
  The optimization begins at the aircraft's actual location and flight state at 08:47 UTC on January 2, 2024, based on ADS-B data.
- Terminal conditions
  The trajectory must terminate precisely at the RNAV IF (Initial Fix) waypoint of the selected diversion airport. This constraint includes:

$$h_{\text{final}} = h_{\text{IF}}, V_{\text{final}} = V_{\text{BADA}}(h_{\text{IF}}) \tag{17}$$

  - h_{IF}: Altitude specified in Japan AIP for the RNAV IF of the respective airport

(typically 3000–3500 ft)

- $V_{\mathrm{BADA}}(h)$: Clean configuration airspeed derived from EUROCONTROL BADA 3.15 PTF, converted to true airspeed

These ensure smooth integration into the Standard Terminal Arrival Route (STAR) and compliance with procedural descent.

- Climb/Descent slope constraint

  The rate of climb or descent between adjacent trajectory points is restricted to feasible bounds defined by aircraft performance limits:

$$\left|\frac{h_{k+1}-h_k}{\Delta t}\right| \leq \text{Climb/Descent Rate}_{\max} \tag{18}$$

- Thrust feasibility

  At each point, the required thrust T_k must satisfy:

$$T_k \leq T_{max}(h_k) \quad (\text{BADA} - \text{defined altitude0dependent maximum}) \tag{19}$$

- Aerodynamic validity

  Aerodynamic parameters such as drag, lift, and fuel flow must be computed within the valid operating envelope defined in the BADA OPF/PTF files. For example:

$$D_k = \frac{1}{2}\rho_k V_{TAS,k}^2 S(C_{D0} + C_{D2}C_L^2) \tag{20}$$

- Recursive mass update

  The aircraft's mass is updated step-by-step to reflect fuel burn:

$$m_{k+1} = m_k - \dot{m_{k,f}} \cdot \Delta t \tag{21}$$

  This ensures thrust and drag calculations reflect the true aircraft weight at each trajectory segment.

- Fuel budget feasibility

  The cumulative fuel used along the trajectory must not exceed the remaining fuel estimated by the Fuel Estimation Tool based on ADS-B trajectory up to 08:47 UTC. If the required fuel exceeds the remaining fuel, the scenario is flagged as infeasible.

Convergence and termination criteria

The trajectory optimization problem is solved using a Sequential Quadratic Programming (SQP) algorithm, precompiled into scenario-specific MEX functions. The solver stops the iteration under any of the following conditions:

- Objective stability
  The change in cost function between successive iterations falls below a threshold:

$$\dot{\Delta J} < 10^{-4}$$

- Maximum iteration count
  The iteration count reaches the upper bound (typically between 100 and 200).
- Gradient tolerance
  The norm of the gradient vector drops below a minimum threshold, indicating local optimality.

These convergence criteria are internally handled within the main_MSDP_Function_mex_nxXX.mexw64 solver. For validation and diagnostics, the optimization process optionally logs residuals, gradient norms, and iteration history.

3.5-7 Output Results

Upon completion of the optimization process, the results for each diverting scenario are saved in individual .mat files. These output files are subsequently used for quantitative comparison against AI-based diversion decisions and for assessing physical feasibility. The main output variables are summarized as follows:

Table 3-11 Key Result of Flight Trajectory Optimizer

Output Variable	Unit	Description
opt_lat, opt_lon, opt_hp	-	Optimized trajectory: latitude, longitude, altitude
opt_time	s	Time elapsed at each point
opt_mass	kg	Aircraft mass over time
opt_FF	kg/s	Fuel flow at each point
opt_fuel	kg	Cumulative fuel burned
Fuel_consumption	kg	Total fuel consumption
Flight_time	s	Total flight time

A complete list of output variables and data structures used in the flight trajectory optimizer module is provided in Appendix C.

The results form the physical baseline for validating whether AI-generated diversion choices are executable under realistic conditions and constraints.

3.6 Comparison methodology

To assess the effectiveness of the proposed DQN-based diversion optimization system, this study conducts a structured comparison between actual pilot decisions and the AI-generated outcomes. Two separate perspectives are analyzed: a fuel-efficiency-focused model and a passenger-convenience-oriented model. Each output is compared against actual diversion records collected from January 2nd, 2024.

The comparison is conducted using four core metrics, as summarized in Table 3-12.

Table 3-12 Comparison Strategy Overview

Metric Name	Description	Applicable Models	Comparison Method
Airport Distribution	Visual comparison of selected airports across all flights	All models	Map-based distribution + summary tables
Airport Matching Classification	Categorization into matched diversions, different diversions, and return-to-origin mismatches	Fuel-efficiency DQN / Passenger-convenience DQN	Case-by-case classification and count
Diversion Time	Time from diversion point to the selected airport	All models	Both average and total time compared
Fuel Consumption	Estimated fuel burned from diversion point to the selected airport	models	Both average and total fuel compared

Flight distributions are visualized on an actual map, with bar graphs overlaid at each airport location to intuitively highlight spatial clustering patterns. This visual analysis helps assess the geographical consistency of decision-making for the actual pilots, the fuel-efficiency-focused model, and the passenger-oriented model.

The airport matching classification identifies the extent to which each DQN model replicated actual diversion choices. All diversion cases are assigned to one of three mutually exclusive categories: (1) matched diversions, where the model chose the same diversion airport as the actual flight; (2) different diversions, where both the actual and model outcomes involved diversion but to different airports; and (3) return-to-origin vs. diversion mismatches, where the actual flight returned to its departure

airport but the model diverted elsewhere. This classification allows for a more nuanced understanding of agreement and disagreement patterns between human and AI decisions.

Diversion time is analyzed both as an average per flight and as a total cumulative value to capture both individual decision realism and overall system efficiency. Similarly, fuel consumption is evaluated using both average and total values. All fuel estimates are based on trajectory optimization outputs or historical trajectory data combined with aircraft performance models (BADA) and meteorological datasets (ERA5 for actual cases, JMA GSM2 for DQN cases).

Certain dimensions—such as post-diversion operational delays or pilot decision latency—were excluded due to data unavailability. Nevertheless, the DQN system generated decisions within under 10 seconds per aircraft, indicating its feasibility for real-time use in emergency diverting scenarios.

Chapter 4
Results

4. Results

4.1 DQN training performance analysis

To validate the learning efficiency and stability of both Deep Q-Network (DQN) variants, the author tracked total episodic reward during 2000 training epochs. The results are visualized in Figures 4-1 and 4-2.

4.1-1 Fuel-efficiency-focused DQN

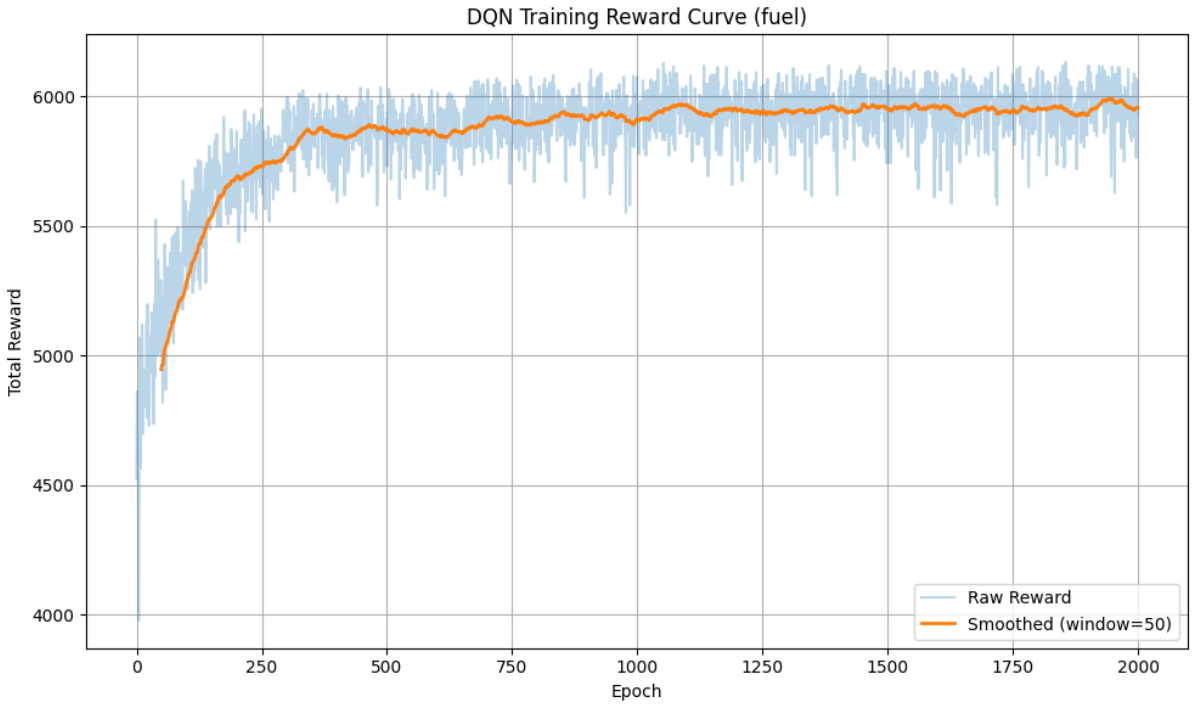


Figure 4-1 Training Reward Curve of Fuel-Efficiency-Focused DQN

Figure 4-1 presents the training reward curve for the fuel-efficiency-focused DQN model over 2,000 epochs. The model demonstrates a rapid increase in total reward within the first 300 epochs, indicating swift policy improvement in the early stages of training. The smoothed reward trend, shown as an orange line using a moving average with a window size of 50, begins to stabilize around epoch 500 and continues to exhibit gradual improvements until reaching convergence near epoch 1800. By the end of training, the total reward stabilizes around 6,000, reflecting the model's ability to consistently identify effective policies. After approximately epoch 1000, the variance in the raw rewards narrows significantly, further confirming the model's convergence.

$$Reward = (w_1 \cdot \text{dist_divert_score} + w_2 \cdot \text{dist_HND_score} + w_3 \cdot \text{origin_score} + w_4 \cdot \text{fuel_score}) \times 200 \quad (22)$$

Where

$$w_1 = 0.2, w_2 = 0.2, w_3 = 0.2, w_4 = 0.4$$

In this fuel-efficiency-focused configuration, the relatively higher value of w_4 prioritizes minimizing fuel consumption, while the other terms ensure operational feasibility by considering diversion distance, distance to Haneda, and return-to-origin factors.

Overall, the reward trend suggests that the DQN model has successfully learned to minimize fuel consumption while adhering to operational constraints such as fuel sufficiency and slot availability.

4.1-2 Passenger-convenience-focused DQN

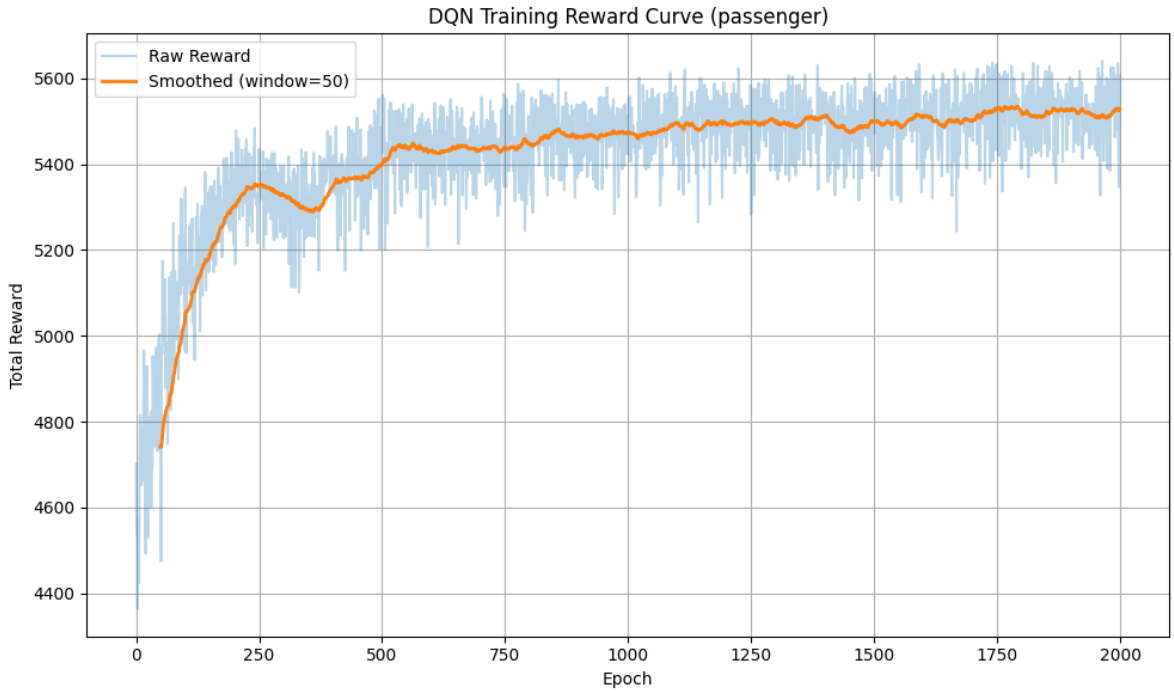


Figure 4-2 Training Reward Curve of Passenger-Convenience-Focused DQN

Figure 4-2 presents the training reward curve for the passenger-convenience-focused DQN model. The overall learning trend exhibits a structure similar to the fuel-efficiency model, with rapid initial improvement and convergence occurring around epoch 600. The smoothed reward curve (orange line), calculated with a moving average window of 50, shows a steady and consistent rise followed by stabilization. However, the final reward plateau—approximately 5500—is lower than that of the fuel-focused model. This is expected, as the passenger-oriented reward function includes additional penalty terms related to transit time, number of transfers, and monetary cost, making optimization inherently more complex. Despite the increased difficulty in balancing multiple objectives, the model converges reliably and does not exhibit signs of overfitting, demonstrating that the composite reward structure is effectively learnable by the DQN.

The reward function for this passenger-focused configuration is defined as:

$$Reward = (w_1 \cdot \text{dist_divert_score} + w_2 \cdot \text{dist_HND_score} + w_3 \cdot \text{origin_score} + w_4 \cdot \text{fuel_score}) \times 200 + PassengerBonus \quad (23)$$

$w_1 = 0.2, w_2 = 0.5, w_3 = 0.2, w_4 = 0.1$, and:

$$\text{PassengerBonus} = (1 - \text{Discomfort}) \times 10 \quad (24)$$

The Discomfort metric aggregates passenger inconvenience factors such as total transit time, number of transfers, and monetary cost from the diversion airport to Haneda. This additional term incentivizes the selection of airports that minimize passenger disruption, even if they are not the most fuel-efficient.

4.2 Diversion airport distribution

To evaluate whether DQN policies yield reasonable and effective diversion decisions, we compare the geographic distribution of diverting results in three scenarios using bar chart maps:

Each map uses:

- Red bars: Number of aircraft diverted to alternative airports
- Yellow bars: Number of aircraft that returned to origin airport

4.2-1 Actual decisions

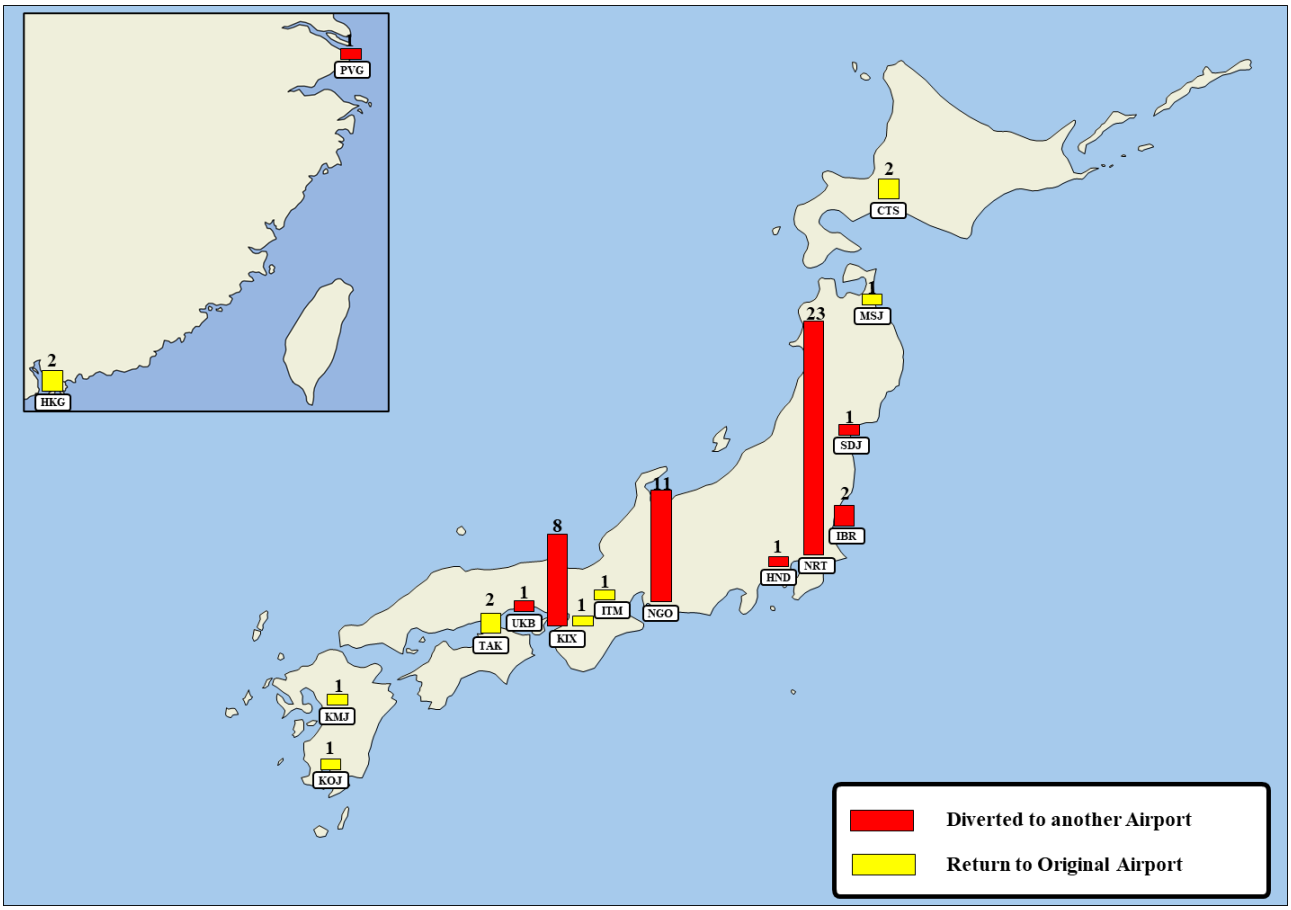


Figure 4-3 Actual Diversions from January 2,2024

The actual diversion records from January 2nd, 2024, reveal a highly concentrated diverting pattern. A total of 23 flights were diverted to Narita International Airport (NRT), which is geographically closest to Haneda (HND), indicating a strong tendency to select the most proximate major airport. Following Narita, some diverting occurred to key hub airports such as Kansai (KIX, 8 flights) and Nagoya (NGO, 11 flights), though to a lesser extent. These choices suggest that pilots and operators may have prioritized airport familiarity, size, or availability of open landing slots during disruption scenarios. Conversely, regional or secondary airports such as Sendai (SDJ), Ibaraki (IBR), and Kobe (UKB) — all within operational range — were rarely utilized, with only 1 to 2 aircraft diverted to each. This underutilization implies that, despite being geographically and operationally viable, these

airports may have been overlooked due to institutional preference, perceived risk, or logistical uncertainty.

4.2-2 DQN fuel model diversions

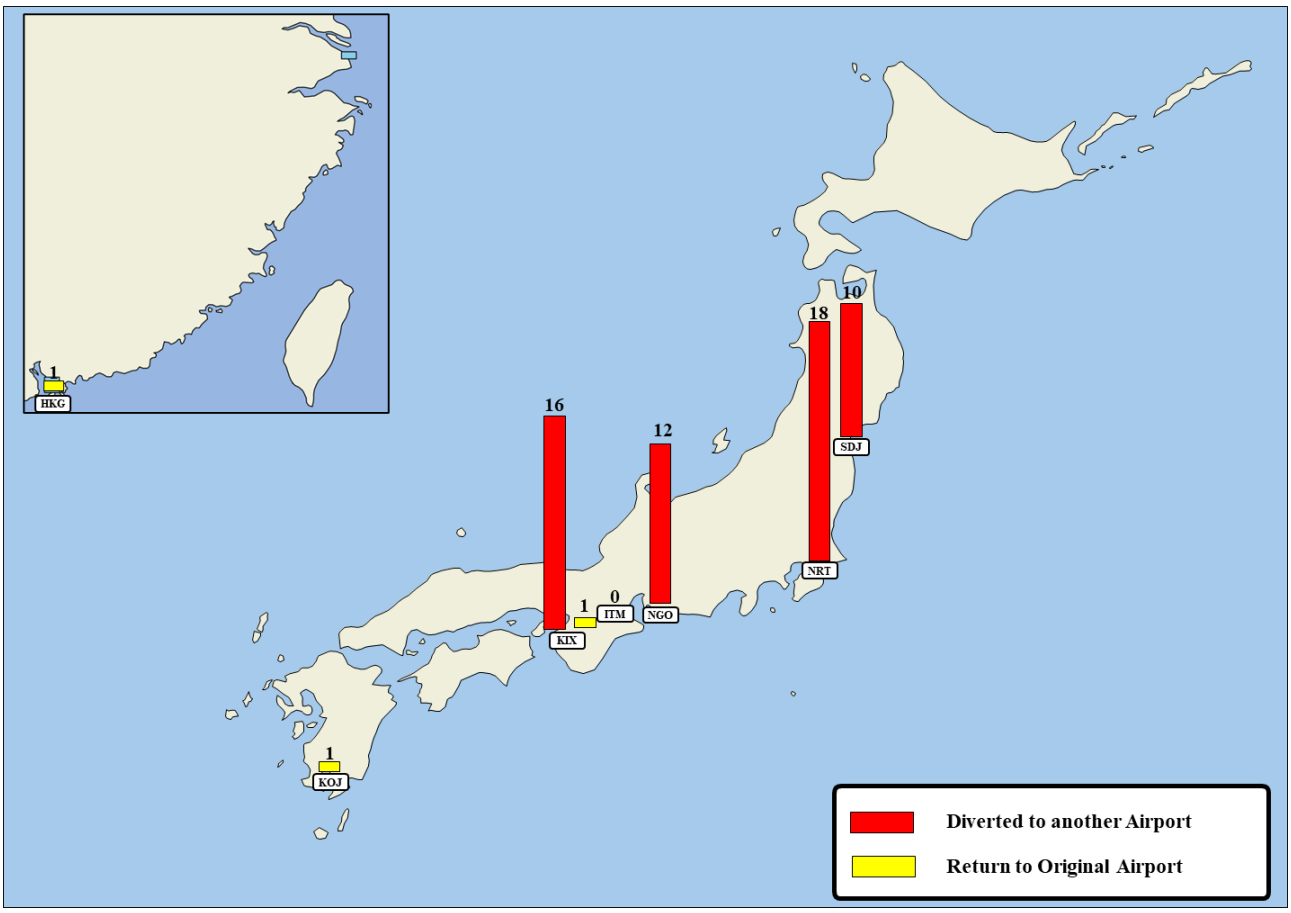


Figure 4-4 DQN (Fuel-Efficiency-Focused) Decisions

The diversion pattern produced by the DQN trained with a fuel-efficiency-focused reward function shows a more balanced and distributed decision structure. In this simulation, a significant number of flights were diverted to Kansai International Airport (KIX, 16 flights) and Nagoya (NGO, 12 flights). These airports were likely selected due to their lower expected fuel consumption compared to more distant or congested alternatives. While Narita (NRT) was still chosen 18 times, it did not dominate the distribution as it did in the actual data, suggesting that the DQN model was able to mitigate airport selection bias and diversify airport usage more effectively. Additionally, the model appears more willing to utilize airports located further from Tokyo, provided the overall fuel profile was favorable — a clear indication of its reward-driven logic.

4.2-3 DQN passenger model diversions

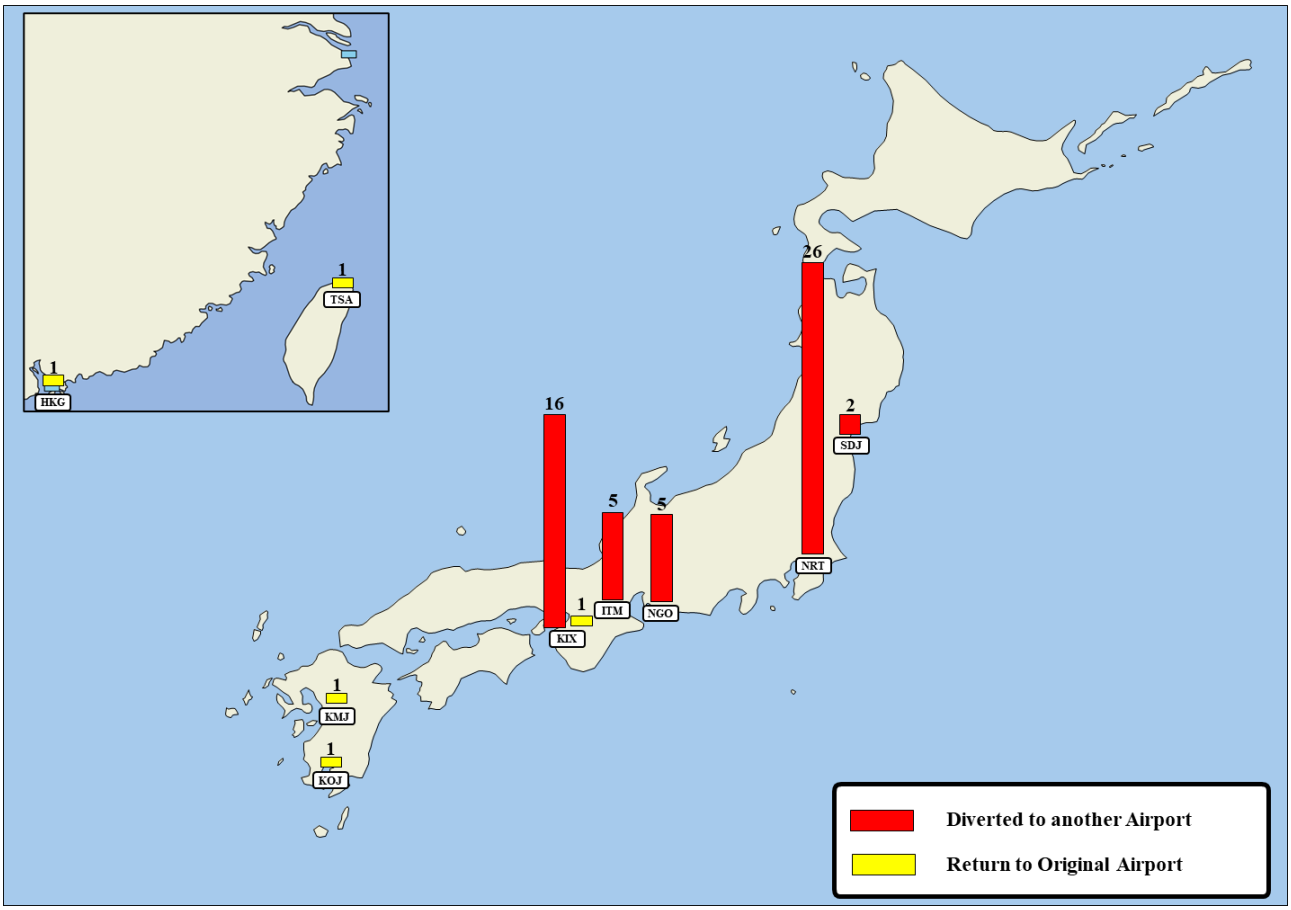


Figure 4-5 DQN (Passenger-Convenience-Focused) Decisions

The DQN model trained with a passenger-convenience-oriented reward function displayed a diversion pattern that closely resembled actual decision behavior, but with important distinctions. Narita (NRT) remained the most frequently selected airport, receiving 27 diverted flights, reinforcing its advantage in terms of accessibility and ground transportation to Tokyo. However, the passenger-oriented model also showed a moderate distribution across other accessible hubs — including Kansai (KIX, 16 flights) and Sendai (SDJ, 2 flights) — reflecting consideration of multi-factor penalties such as total travel time, number of transfers, and transit cost from the diversion airport to the original destination. Additionally, Itami (ITM) and Nagoya (NGO) were each selected 5 times, further indicating that the model learned to balance accessibility with geographic and operational constraints. In contrast to the fuel-based model, this behavior suggests a goal-specific prioritization of post-landing passenger experience over flight operational efficiency.

4.2-4 Tabular comparison of diversion decisions

To quantify differences in airport usage, Table 4-1 presents the count of diverted aircraft per airport in each of the three systems.

Table 4-1 Diversion Count by Airport

Category	Airport	Actual result		DQN (Fuel)		DQN (Passenger)	
-	-	Diverted	Returned	Diverted	Returned	Diverted	Returned
Selected candidate airport	Narita (NRT)	23	0	18	0	26	0
	Kansai (KIX)	8	1	16	1	16	1
	Nagoya (NGO)	11	0	12	0	5	0
	Sendai (SDJ)	1	0	10	0	2	0
	Itami (ITM)	0	1	0	0	5	0
Non-selected Domestic airport	Haneda (HND)	1	0	0	0	0	0
	Ibaraki (IBR)	2	0	0	0	0	0
	Kobe (UKB)	1	0	0	0	0	0
	Takamatsu (TAK)	0	2	0	0	0	0
	Misawa (MSJ)	0	1	0	0	0	0
	Shin-Chitose (CTS)	0	2	0	0	0	0
	Kumamoto (KMJ)	0	1	0	0	0	1
	Kagoshima (KOJ)	0	1	0	1	0	1
Non-selected International airport	Shanghai (PVG)	1	0	0	0	0	0
	Taipei (TSA)	0	0	0	0	0	1
	Hongkong (HKG)	0	2	0	1	0	1
-	Total	48	11	56	3	54	5
		59		59		59	

4.2-5 Interpretation and insights

To assess the strategic characteristics and realism of the diversion decisions made by the DQN-based models, a comparative evaluation was conducted among actual outcomes, the fuel-efficiency-focused DQN, and the passenger-convenience-oriented DQN. The analysis spans several key dimensions including airport selection trends, return behaviors, consistency of decisions, and decision latency.

Table 4-2 Operational Characteristics of Diversion Decisions under Real and DQN Policies

Metric	Real Flights	DQN (Fuel)	DQN (Passenger)
Most Used Airport	NRT (23 flights)	NRT (18 flights)	NRT (26 flights)
Flights to the Airports except 5 selected Airports[1]	14	2	4
Return to Origin Cases	11	3	5
Decision Consistency	Medium	High	High
Decision time	Unknown	< 10 sec (excluding trajectory optimization)	< 10 sec (excluding trajectory optimization)

In terms of airport concentration, the actual data shows that Narita International Airport (NRT) was the most frequently chosen diversion destination, with 23 flights diverted there. This selection likely reflects NRT's proximity to Haneda Airport, extensive ground handling capacity, and high familiarity among flight crews and dispatch centers. In contrast, the fuel-efficiency-oriented DQN model exhibited a more distributed behavior, most notably redirecting 16 flights to Kansai International Airport (KIX). This outcome may not solely be attributed to fuel efficiency considerations; rather, it is likely influenced by the capacity-related term in the reward function, which can penalize overly congested airports such as NRT, as well as the spatial distribution of aircraft positions at the time of the accident. In particular, for several flights in the dataset, KIX was geographically closer than NRT at the diversion decision point, making it a more favorable option under the model's optimization criteria. The passenger-oriented DQN, however, displayed a return to strong NRT preference, diverting 26 flights to the airport. This reflects a prioritization of passenger accessibility and transit convenience, which aligns closely with actual tendencies.

[1] 5 selected airports are Narita (NRT), Kansai (KIX), Sendai (SDJ), Itami (ITM), Nagoya (NGO)

When considering the number of airports used outside the five main candidates (NRT, KIX, NGO, SDJ, ITM), the actual cases involved 14 distinct diversion airports, indicating a broader, potentially less centralized decision-making pattern. This diversity includes several regional and international airports, and may reflect both situational flexibility and the absence of uniform diverting protocols. In contrast, the DQN models operated with greater selectivity: the fuel-efficiency model utilized only 2 distinct airports, and the passenger-convenience model extended to 4. This reduction in airport diversity highlights the reinforcement learning system's convergence toward consistent high-reward decisions, driven by well-defined optimization goals.

Return-to-origin behavior was another observable pattern across all scenarios. The actual data included 11 such instances, suggesting that under certain conditions, returning to the departure airport was considered both feasible and preferable. The DQN models retained this option but used it more carefully: 3 returns were recorded in the fuel model and 5 in the passenger model. This behavior illustrates that while the models retained flexibility to identify return routes as valid alternatives, they also tended to impose stricter evaluation criteria, possibly due to implicit or explicit penalties embedded in the reward structure.

Decision consistency further differentiates the models from actual practice. Human-operated diversions exhibited medium consistency, likely influenced by fragmented situational awareness, real-time communication challenges, and subjective judgment. The DQN systems, by contrast, produced highly consistent outputs, responding to similar operational conditions with decisions that follow a clear set of rules and can be reproduced reliably. This consistency underscores the operational reliability of algorithmic diverting during widespread disruptions and enhances scalability in deployment scenarios.

Finally, regarding decision time, the DQN systems demonstrated significantly faster decision-making capabilities. With pre-trained models and a discrete action space, the agent completed each decision cycle in under 10 seconds per aircraft. It is important to note that this measurement reflects only the policy inference time for selecting a diversion airport and does not include the time required for subsequent trajectory optimization. In practice, trajectory optimization — which involves computing a fuel-efficient or time-efficient path from the aircraft’s current position to the selected diversion airport — can add several seconds to minutes depending on factors such as path grid resolution, meteorological input complexity, and available computational resources. While the exact latency of actual diverting decisions is difficult to determine, it typically involves multi-layered coordination among air traffic controllers, airline operations centers, and flight crews, which introduces unavoidable delays.

Overall, the results indicate that DQN-based models are not only capable of replicating actual decision patterns but also enhancing them in measurable ways. The fuel model maximizes remaining fuel reserves, contributing to system-wide efficiency in energy usage, while the passenger-focused model aligns closely with human-centric priorities such as convenience and accessibility. Together, these findings validate the flexibility and effectiveness of reinforcement learning in aviation diverting under emergency conditions.

4.3 Detailed comparison of model outputs with actual diversion decisions

To gain deeper insight into the similarities and differences between the DQN models' outputs and actual diversion decisions, all 59 diversion cases were classified into three mutually exclusive categories:

- Matched Diversions – Cases where the diversion airport selected by the model exactly matched the airport recorded in the actual flight data. Return-to-origin cases are excluded.
- Different Diversions – Cases where both the actual flight and the model diverted to different airports. Neither case involved a return to the departure airport.
- Return-to-Origin vs Diversion Mismatch – Cases where the actual flight returned to its departure airport, but the model diverted to a different airport.

This classification avoids double counting and provides a clearer picture of when and why the model's decisions align or diverge from actual operations.

Table 4-3 Classification of DQN Decision Outcomes Compared to Actual Diversions

Category	Definition	Fuel DQN	Example flights (Fuel)	Passenger DQN	Example flights (Passenger)
Matched Diversions	Same diversion airport as actual flight	19	6J62, AA175, BR190, JL128…	26	AA175, HA457, JL128, JL166…
Different Diversions	Different diversion airport than actual flight (both diverted)	31	6J96, AF274, CZ3085, JL26…	25	6J62, 6J96, AF274, BC618…
Return-to-Origin vs Diversion Mismatch	Actual returned to origin but model diverted elsewhere	9	BC618, JL26, NH34. NH72…	8	BC618, JL518, JL26, NH648

Interpretation

The fuel-efficiency-focused DQN matched the actual diversion airport in 19 out of 59 cases (32.2%), while the passenger-convenience-focused model achieved 26 matches (44.1%). Although the passenger model achieved a higher exact match count, this does not necessarily imply superior operational performance, as each model optimizes for different priorities.

The fuel model produced more Different Diversions (31 vs 25), reflecting its tendency to explore geographically closer or fuel-efficient alternatives that differ from actual patterns. For example, in several cases where flights diverted to Narita in reality, the fuel model selected Kansai (KIX) or Nagoya (NGO) due to lower estimated fuel consumption from the aircraft's position at the time of the accident.

Return-to-Origin vs Diversion Mismatch occurred in 9 fuel-model cases and 8 passenger-model cases. These discrepancies indicate situations where the models chose to divert rather than return to origin, potentially due to reward penalties for long return flights or limited benefit in terms of the optimization objectives.

Overall, this breakdown shows that while both models replicate actual decisions in a substantial number of cases, their divergences are often systematic rather than random, reflecting the underlying reward design and the operational priorities encoded within each model.

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4.4 Average and total time comparison

To assess the time efficiency of diversion decisions, we evaluated both the average diversion time per flight and the total cumulative diversion time across all 59 flights. These dual indicators reflect both the realism of individual diverting paths and the operational burden on the entire air traffic system.

Table 4-4 Comparison of Diversion Time Across Models

Model	Average Diversion Time (min:sec)	Total Diversion Time (hr:min:sec)
Actual flights	90:54	89:23:46
Fuel-Efficiency DQN	44:36	43:52:11
Passenger-Convenience DQN	44:09	43:24:57

The results clearly highlight the advantage of DQN-based models in minimizing diverting time. Compared to the actual data, which averaged over 90 minutes per flight, both DQN systems achieved nearly 50% reduction in average diversion duration, completing diverting in approximately 44 minutes per aircraft.

In terms of total time, the fuel model saved over 45 hours, and the passenger model saved nearly 46 hours compared to historical cases. This gain is attributed to the trajectory optimization engine used in the DQN environment, which computes near-optimal paths from aircraft position to alternative airports, considering current weather and fuel constraints.

By contrast, actual diversions are often shaped by procedural detours, holding patterns, and manual coordination, resulting in longer and less efficient paths.

This time-based comparison underscores the system-wide efficiency and responsiveness of reinforcement learning models when integrated with data-driven optimization modules.

4.5 Average and total fuel comparison

To evaluate the fuel efficiency of diversion decisions, both the average fuel consumption per flight and the total cumulative fuel consumption across all 59 flights were analyzed. All values represent estimated consumption rather than measured data. For the real flights, estimates were obtained by applying the Fuel Estimation Tool to recorded trajectories, using BADA aircraft performance profiles and ERA5 reanalysis meteorological data. For the DQN-based models, consumption was estimated from optimized diversion routes generated by the trajectory optimization module, using BADA performance data in conjunction with Japan Meteorological Agency meteorological datasets.

Table 4-5 Comparison of Diversion Fuel Across Models

Model	Average Fuel (kg)	Total Fuel (kg)
Actual flight	7,709.68	454,871.37
Fuel-Efficiency DQN	2,413.24	142,381.04
Passenger-Convenience DQN	2,455.14	144,853.22

The results demonstrate a significant reduction in estimated fuel consumption for both DQN-based models compared to actual diversions. While the actual diversions consumed an average of 7,709.68 kg per flight, the fuel-focused DQN reduced this to 2,413.24 kg, and the passenger-focused DQN to 2,455.14 kg — a decrease of approximately 68–69%. In total, this corresponds to a reduction of over 300,000 kg of fuel across the dataset. The slightly lower consumption of the fuel-focused model reflects its optimization objective, which prioritizes airports and routes that minimize fuel burn, whereas the passenger-focused model balances this with accessibility considerations.

These fuel savings are primarily the result of the trajectory optimization process within the DQN framework, which calculates near-optimal paths from the aircraft's position at the diversion decision point to the chosen airport, incorporating wind, atmospheric conditions, and aircraft performance data. By contrast, actual diversion routes often include procedural deviations, holding patterns, and operational constraints that extend the distance flown and increase fuel consumption. It should be emphasized that these results represent theoretical savings under modeled conditions and do not account for certain actual limitations such as air traffic control interventions, unforeseen weather changes, or ground handling constraints.

Chapter 5
Conclusion

5. Conclusion

This study investigated a reinforcement learning–based approach for optimizing aircraft diversion decisions during sudden airport closures, applying a Deep Q-Network (DQN) framework with two contrasting reward paradigms: fuel-efficiency and passenger-convenience. Using 59 actual flight cases from the January 2nd, 2024 disruption at Haneda Airport, the models were evaluated through quantitative and qualitative comparisons against actual diversion records.

The results show that both DQN variants produced consistent and structured outputs within seconds per aircraft—excluding trajectory optimization—demonstrating a clear advantage in decision speed over the multi-layered coordination required in real operations. As shown in Table 4-4, the fuel-efficiency-focused model reduced average diversion time from 90 minutes (real flights) to 44 minutes per aircraft, saving over 45 hours in total across the dataset, while the passenger-convenience-oriented model achieved similar time savings. In terms of estimated fuel burn (Table 4-5), the fuel-focused model reduced total consumption from 454,871 kg to 142,381 kg (−312,490 kg), with the passenger model at 144,853 kg, reflecting the efficiency of optimized diverting paths.

Operationally, the fuel-efficiency-focused model avoided over-concentration at Narita International Airport (NRT), instead selecting alternatives such as Kansai (KIX) or Nagoya (NGO) when advantageous for fuel or traffic efficiency. The passenger-convenience-oriented model, by contrast, maintained strong preference for NRT, reflecting its prioritization of accessibility, onward travel time, and transit convenience.

These findings highlight that reinforcement learning can be tailored to meet different operational priorities in emergency diverting—be it minimizing fuel burn and congestion or enhancing passenger experience. The consistent decision patterns across similar conditions suggest that the models can serve as reliable decision aids, offering predictable and transparent logic compared to human-influenced variability.

However, several limitations must be acknowledged. All time and fuel values were estimated under modeled conditions, not directly measured from aircraft records. Actual constraints—such as air traffic control interventions, unforeseen weather, or dynamic airport capacity changes—were not incorporated. Furthermore, the network of candidate diversion airports was fixed across all scenarios, potentially limiting adaptability in more complex or evolving operational contexts.

Future work should integrate real-time data streams, dynamically expand or contract the set of candidate airports based on prevailing conditions, and adopt multi-objective reinforcement learning

to balance fuel efficiency, passenger accessibility, airport capacity, and operational cost. By addressing these factors, DQN-based diversion systems could evolve into practical, real-time decision-support tools for airlines and air traffic managers in disruption-prone environments.

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References

[1] Clausen, J., Larsen, A., Larsen, J., & Rezanova, N. J. (2010). Disruption management in the airline industry—Concepts, models and methods. Computers & Operations Research, 37(5), 809–821.

[2] Hassan, L. K., Santos, B. F., & Vink, J. (2021). Airline disruption management: A literature review and practical challenges. Computers & Operations Research, 127, 105137.

[3] EUROCONTROL. (2019). User Manual for the Base of Aircraft Data (BADA), Revision 3.15. Brussels: EUROCONTROL.

[4] Hersbach, H., Bell, B., Berrisford, P., et al. (2020). The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society, 146(730), 1999–2049.

[5] Xu, Y., Zhou, J., Chen, N., & Sridhar, B. (2023). Machine-Learning-Assisted Optimization of Aircraft Trajectories under Realistic Constraints. Journal of Guidance, Control, and Dynamics, 46(9), 1705–1722.

[6] Wang, Z., Vouros, G., Papadopoulos, G., et al. (2022). Review of deep reinforcement learning approaches for air traffic control. Aerospace, 9(6), 294.

[7] Papadopoulos, G., Vouros, G., Bastas, A., Cordero, J. M., & Rodríguez, R. R. (2024). Deep reinforcement learning in service of air traffic controllers to resolve tactical conflicts. Expert Systems with Applications, 242, 121234.

[8] Razzaghi, P., Tabrizian, A., Guo, W., Chen, S., Taye, A., Thompson, E., Bregeon, A., Baheri, A., & Wei, P. (2024). A survey on reinforcement learning in aviation applications. Engineering Applications of Artificial Intelligence, 136(Part A), 108095.

[9] Itoh, E. (2023). Introduction to Air Traffic Management System. Corona Publishing Co, Ltd,

[10] Diversion. (n.d.). SKYbrary Aviation Safety. FAA JO 7110.65

https://skybrary.aero/articles/diversion

[11] Pilanawithana, N., M., Feng. Y., London, K., Zhang, P. (2022). Developing resilience for safety management systems in building repair and maintenance: A conceptual model

[12] Razzaghi, P., Tabrizian, A., Guo, W., Chen, S., Taye, A., Thompson, E., Bregeon, A., Baheri, A., Wei, P. (2024). A Survey on Reinforcement Learning in Aviation Application

[13] Alverez, L., E., Brittain, M., W. (2024). Tradeoffs When Considering Deep Reinforcement Learning for Contingency Management in Advanced Air Mobility

[14] Flightradar24. (n.d.). Flightradar24: Live flight tracker. Retrieved August 4, 2025. https://www.flightradar24.com

[15] Copernicus Climate Change Service. (2017). ERA5 hourly data on pressure levels from 1959 to present [Data set]. Copernicus Climate Data Store (CDS). Retrieved August 4,2025. https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-pressure-levels

[16] Celikel, A., Jelinek, F. (2001). Forecasting Civil Aviation Fuel Burn and Emissions in Europe (Interim Report)

[17] Paszke, A., & Towers, M. (2025, June 16). Reinforcement learning (DQN) tutorial [Tutorial]. PyTorch. https://docs.pytorch.org/tutorials/intermediate/reinforcement_q_learning.html

[18] Miyamoto, Y., Wickramasinghe, N. K., Harada, A., Miyazawa, Y., & Funabiki, K. (2013). Analysis of fuel-efficient airliner flight via dynamic programming trajectory optimization. (pp. 93–98).

[19] Japan Civil Aviation Bureau. (2024). Aeronautical Information Publication (AIP Japan), RNAV STAR Charts. Ministry of Land, Infrastructure, Transport and Tourism. https://aisjapan.mlit.go.jp/

[20] Japan Meteorological Agency (JMA). (2024). Global Spectral Model (GSM) analysis data, single-level fields (GSM2) [Data set]. Retrieved August 4, 2025.

Appendix

Appendix A: DQN Model Hyperparameters

Table A-1 DQN Hyperparameters List

Parameter	Value
Input size	6 (state dimensions)
Output size	5–6 (number of candidate airports)
Hidden layers	Two layers, 128 neurons each
Activation function	ReLU
Optimizer	Adam (learning rate = 0.001)
Discount factor (γ)	0.95
Replay buffer size	10,000 transitions
Batch size	32
ε-schedule (exploration)	$\varepsilon_0 = 1.0 \rightarrow \varepsilon_min = 0.05$ (linear decay)
Loss function	Mean Squared Error (MSE)
Number of training episodes	2000

Appendix B: Passenger Convenience Reward Data Collection from Google Maps

B.1 Public Transportation routes from Diverted Airports to Haneda Airport

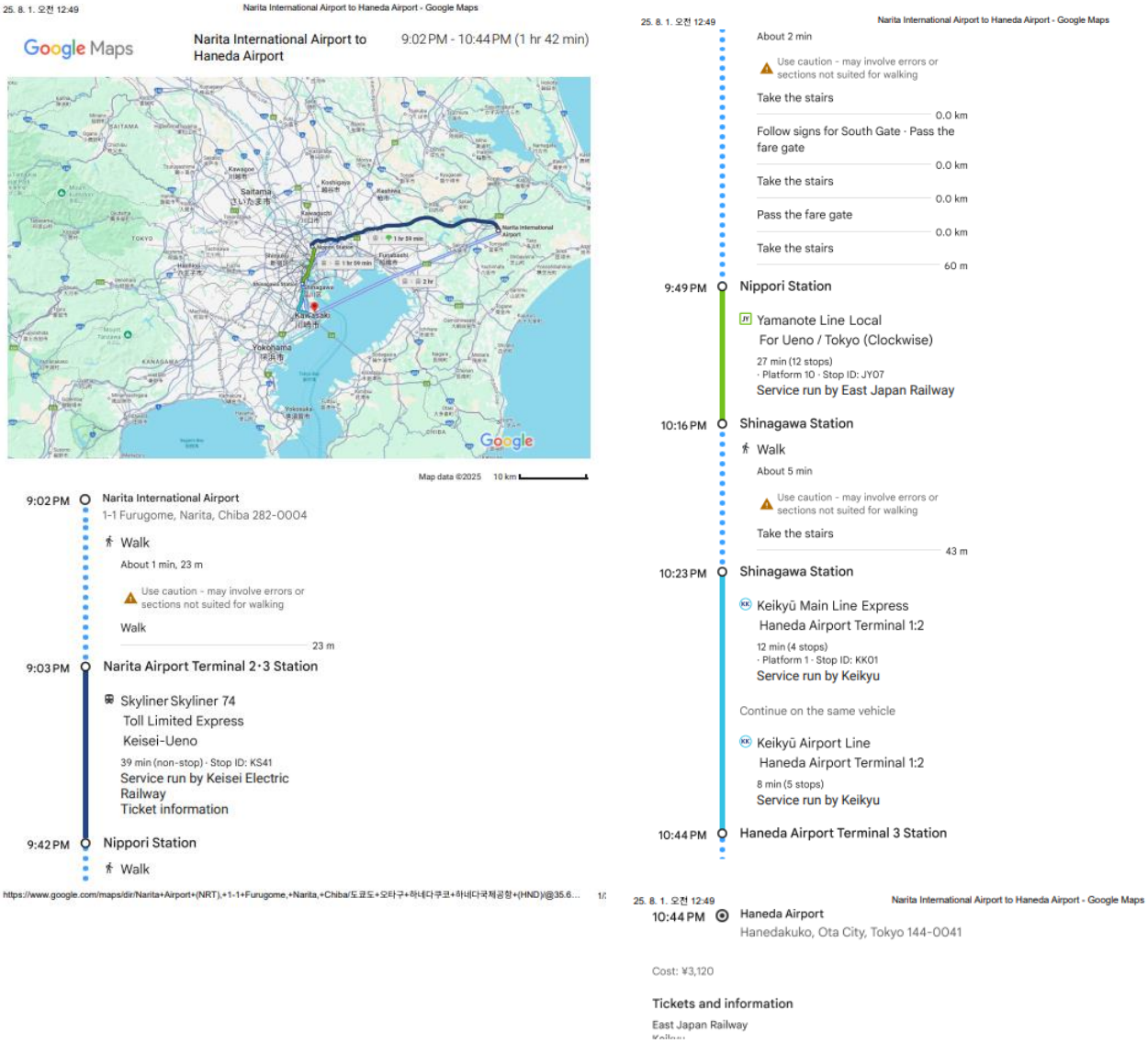

Figure B-1 Public transportation route from Narita International Airport (NRT) to Haneda Airport (HND) as recommended by Google Maps.

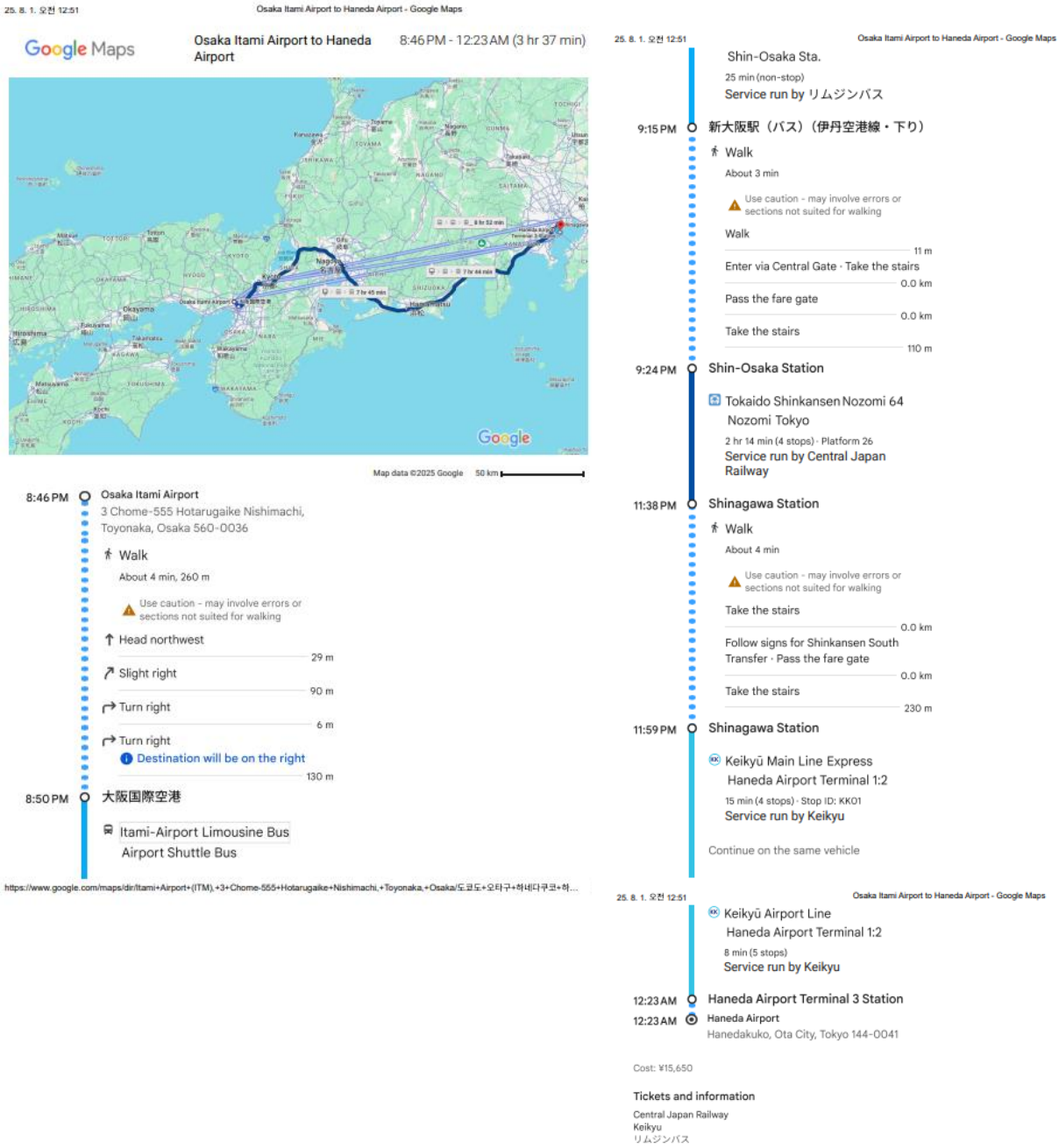

Figure B-2 Public transportation route from Osaka Itami Airport (ITM) to Haneda Airport (HND) as recommended by Google Maps

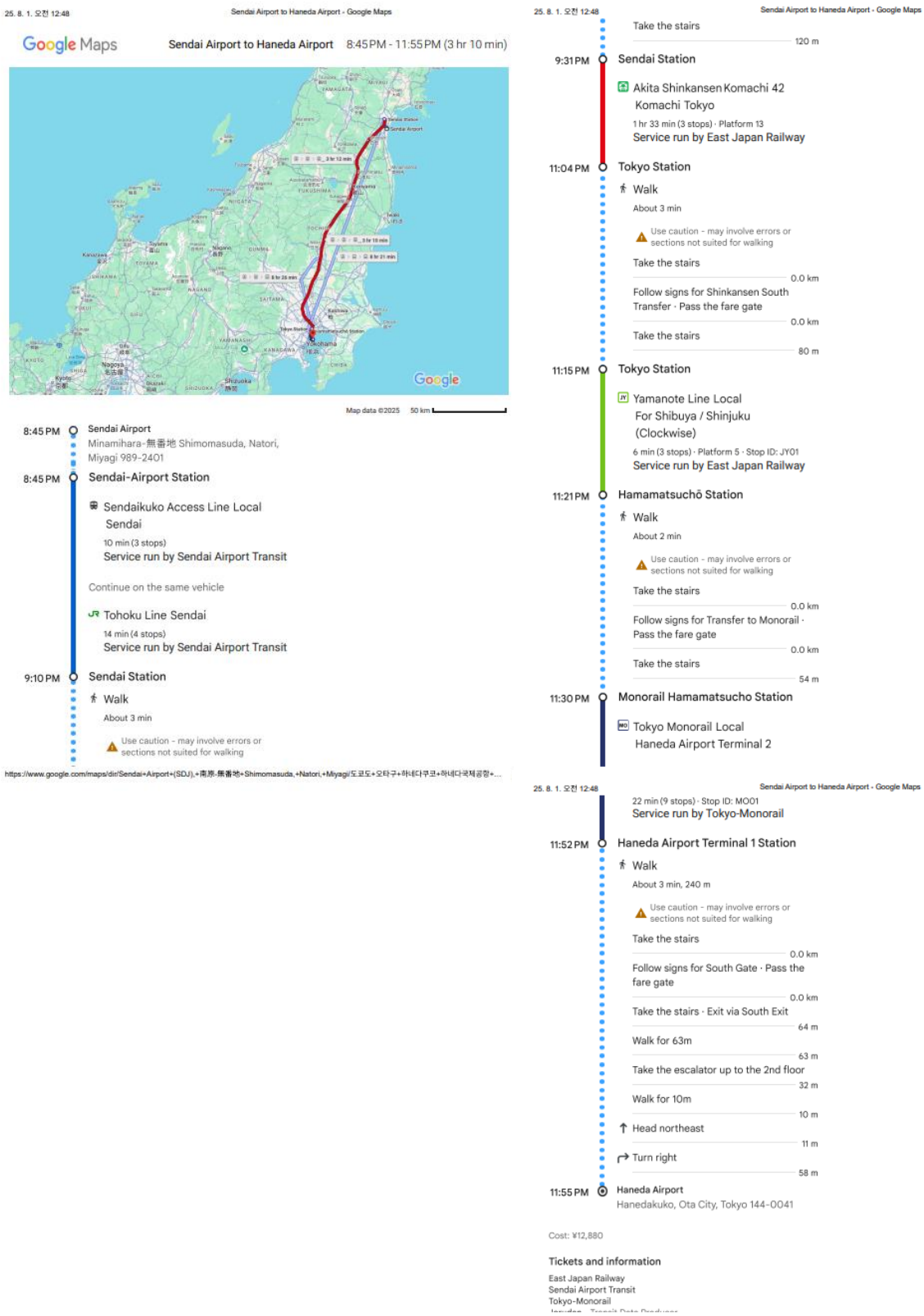

Figure B-3 Public transportation route from Sendai Airport (SDJ) to Haneda Airport (HND) as recommended by Google Maps.

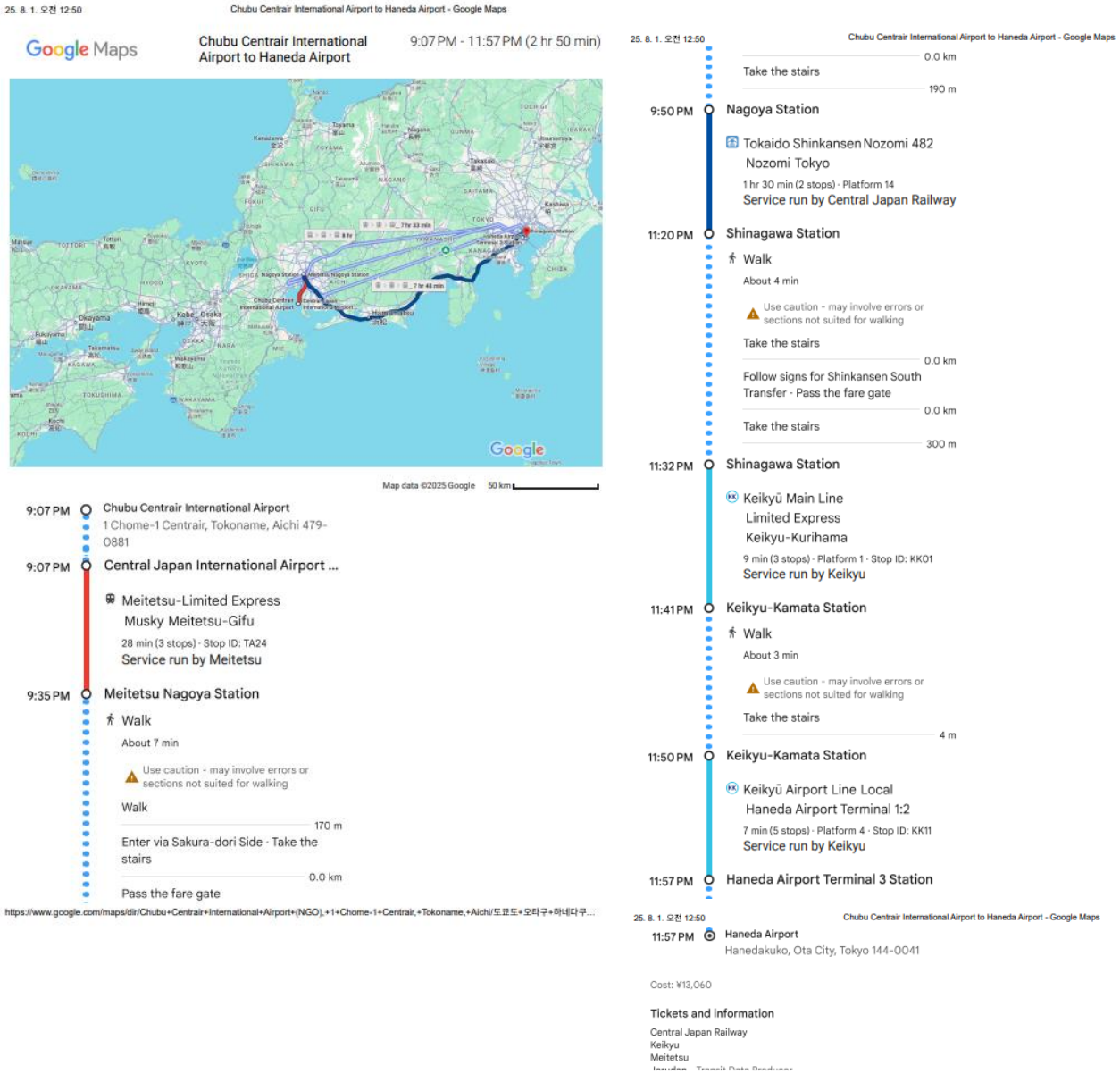

Figure B-4 Public transportation route from Nagoya Airport (NGO) to Haneda Airport (HND) as recommended by Google Maps.

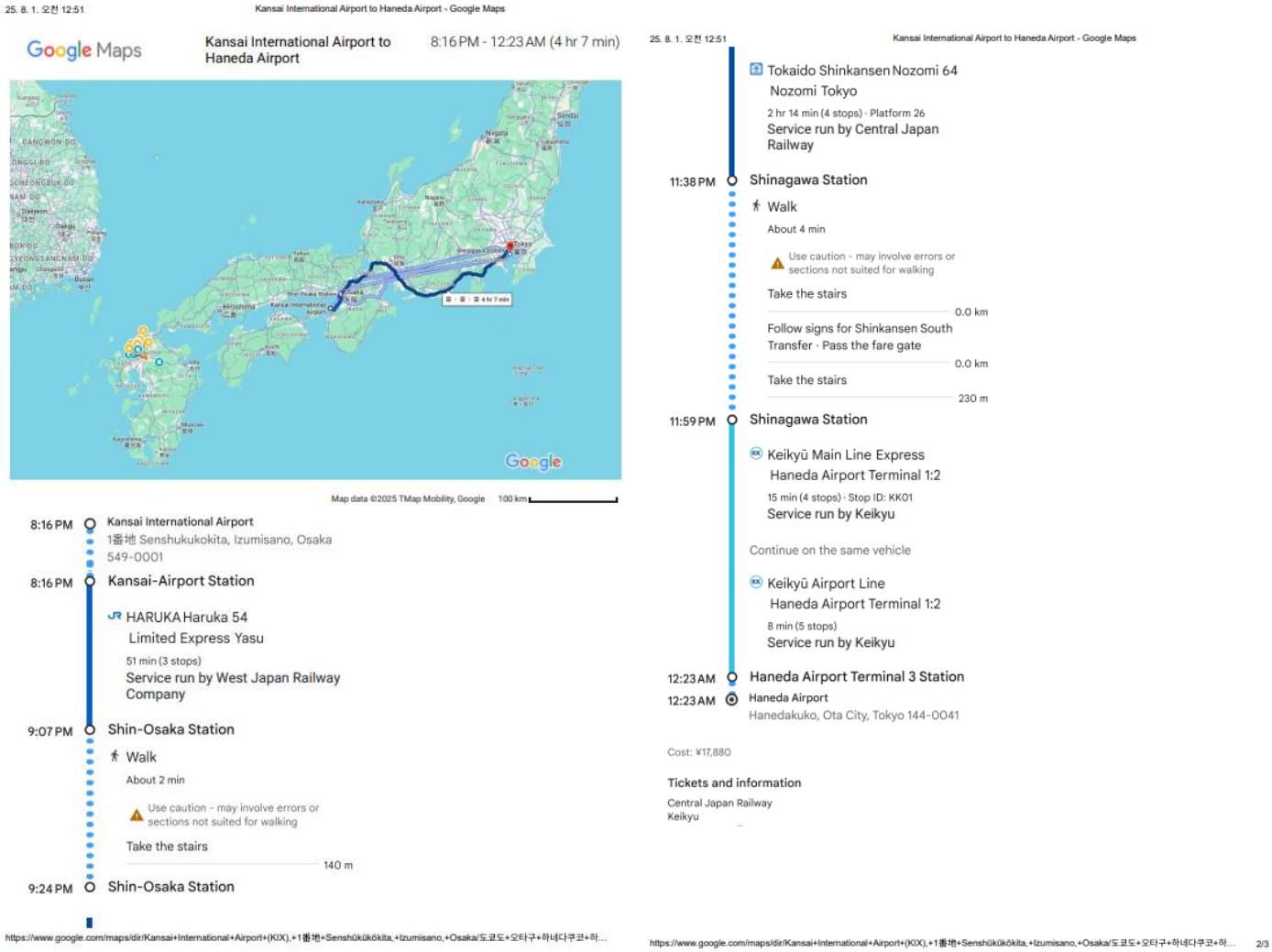

Figure B-5 Public transportation route from Kansai International Airport (KIX) to Haneda Airport (HND) as recommended by Google Maps.

B.2 Summary of Public Transportation Options from Diverted Airports to Haneda Airport

Table B-1 summarized information for public transportation routes from each diverted airport to Haneda Airport, as obtained from Google Maps.

Airport	Total Travel Time (min)	Number of Transfers	Cost (JPY)
NRT	102	3	3,120
ITM	217	3	15,450
SDJ	190	4	12,480
NGO	170	4	12,860
KIX	247	3	17,480

Appendix C: Output Results of Fuel Estimation Tool

Table C-1 TFD Result List of Fuel Estimation Tool

Output Variable	Unit	Description
ACID	-	Unique identifier assigned to each flight scenario (e.g., flight number or internal index), used to match optimization and fuel estimation results.
ACtype	-	ICAO aircraft type code (e.g., B738, A320), used to retrieve the appropriate BADA 3.15 performance model for each flight.
Time_s	s	Total time elapsed up to cutoff time
Fuel_kg	kg	Total fuel burned up to the cutoff time
Dist_km	km	Total distance flown up to cutoff time

Table C-2 Result_est List of Fuel Estimation Tool

Output Variable	Unit	Description
act_time	s	Time series
act_palt	M	Pressure altitude
act_lon	deg	Longitude
act_lat	deg	Latitude
est_rho	kg/m^3	Estimated air density
est_GS	m/s	Estimated ground speed
est_CAS	m/s	Estimated calibrated airspeed
est_TAS	m/s	Estimated true airspeed
est_MACH	-	Estimated Mach number
est_gamma	rad	Flight path angle
est_psia	rad	Heading angle
est_delW	kg/s	Fuel weight flow delta
est_Thr	N	Required thrust
est_Tmax	N	Maximum available thrust
est_CL	-	Lift coefficient
est_CD	-	Drag coefficient
est_L	N	Lift force
est_D	N	Drag force
est_FF	kg/s	Estimated fuel flow
est_mass	kg	Estimated aircraft mass

Appendix D: Output Results of Flight Trajectory Optimizer

Table D-1 Result_opt List of Flight Trajectory Optimizer

Output Variable	Unit	Description
opt_hp	m	Pressure altitude
opt_CAS	m/s	Calibrated Air Speed
opt_lon	deg	Longitude
opt_lat	deg	Latitude
opt_TAS	m/s	True Air Speed
opt_MACH	-	Mach number
opt_delt	-	Engine delta thrust control input
opt_gamma_a	rad	Flight path angle
opt_psi_a	rad	Heading angle
opt_dw	kg/s	Fuel weight flow
opt_FF	kg/s	Fuel flow
opt_T	N	Required thrust
opt_Tmax	N	Maximum available thrust
opt_CL	-	Lift coefficient
opt_CD	-	Drag coefficient
opt_L	N	Lift force
opt_D	N	Drag force
opt_uwind	m/s	Wind component (u-direction)
opt_vwind	m/s	Wind component (v-direction)
opt_tmpc	°C	Temperature
opt_rho	kg/m^3	Air density
opt_p	Pa	Air pressure
opt_hgeo	m	Geopotential height
opt_a	m/s	Speed of sound
opt_time	s	Time at each segment
opt_fuel	kg	Cumulative fuel burn
opt_mass	kg	Aircraft mass
Fuel_consumption	kg	Total fuel burned
Flight_time	S	Total flight duration
dw1, dw2	-	Internal fuel computation variables
Thr1, Thr2	-	Internal thrust computation variables
Tmax_1, Tmax_2	-	Maximum thrust parameters
flight_id	-	Flight identifier